

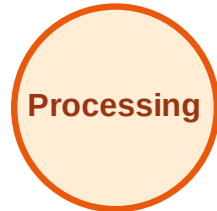
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



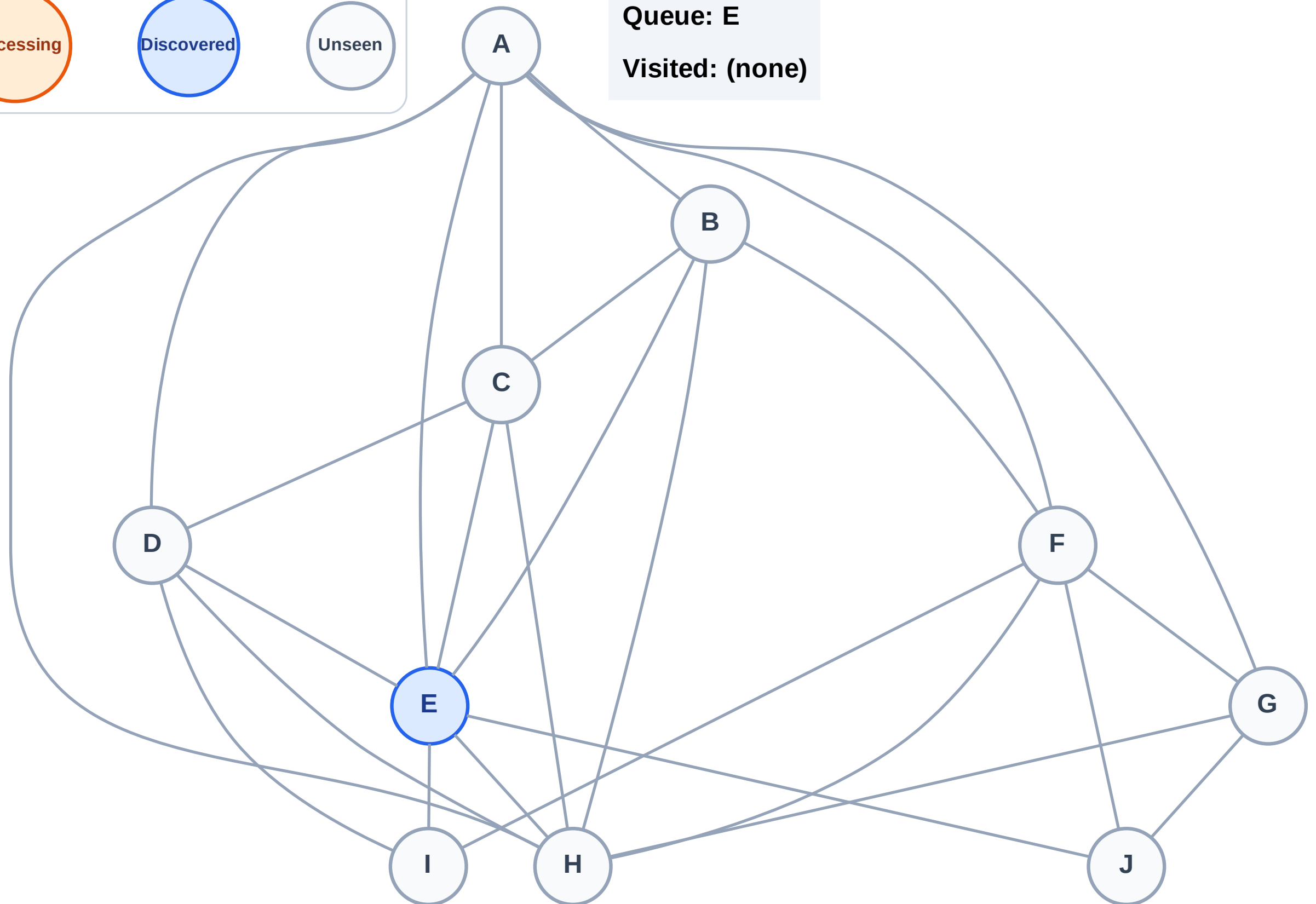
Discovered



Unseen

Queue: E

Visited: (none)



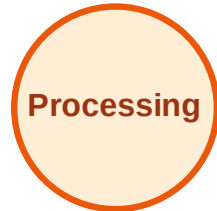
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



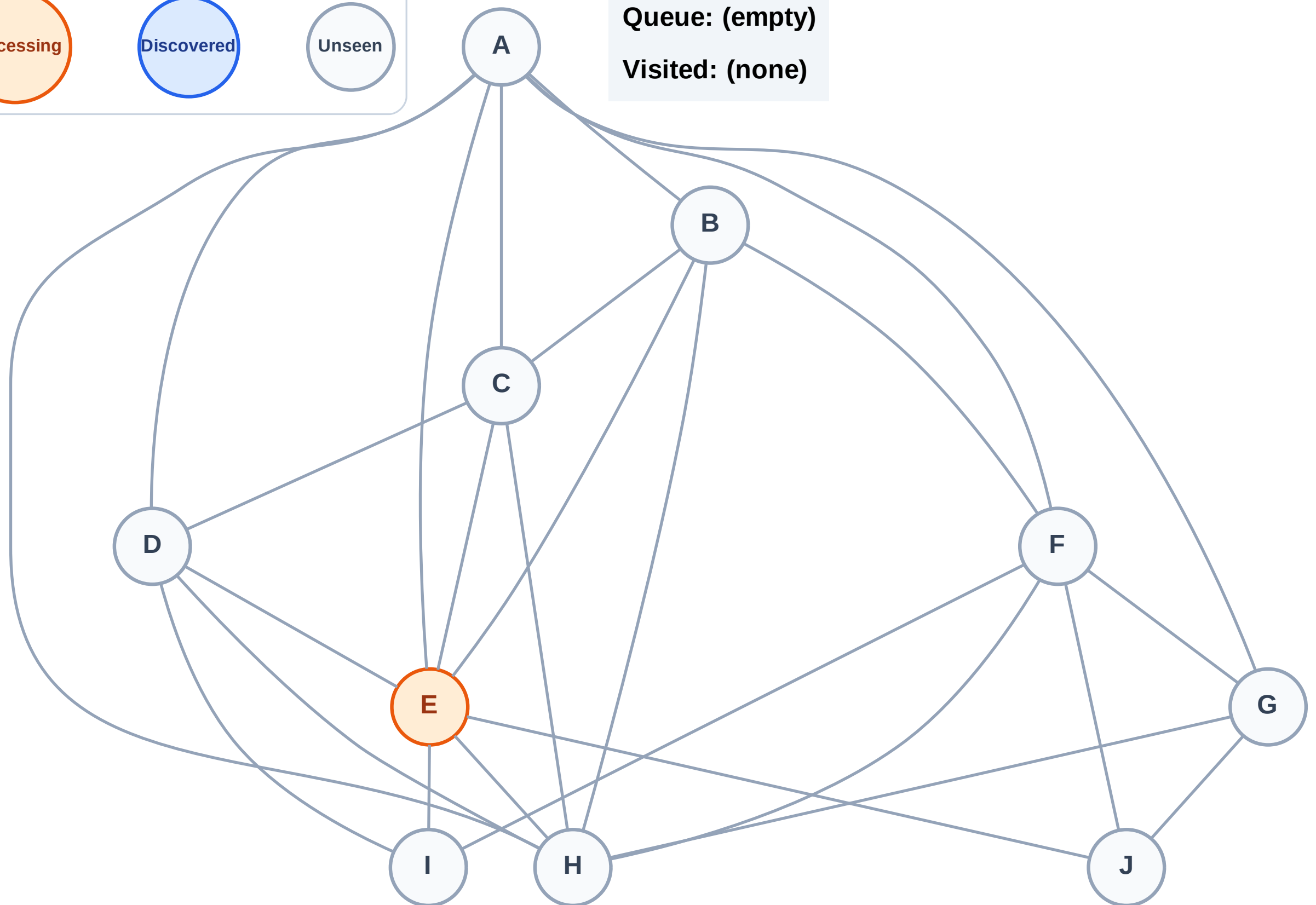
Discovered



Unseen

Queue: (empty)

Visited: (none)



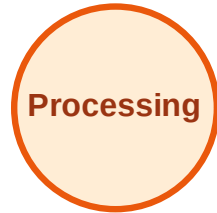
BFS Traversal

Step 3: Discover A, B, C, D, H, I, J from E

Legend



Complete



Processing



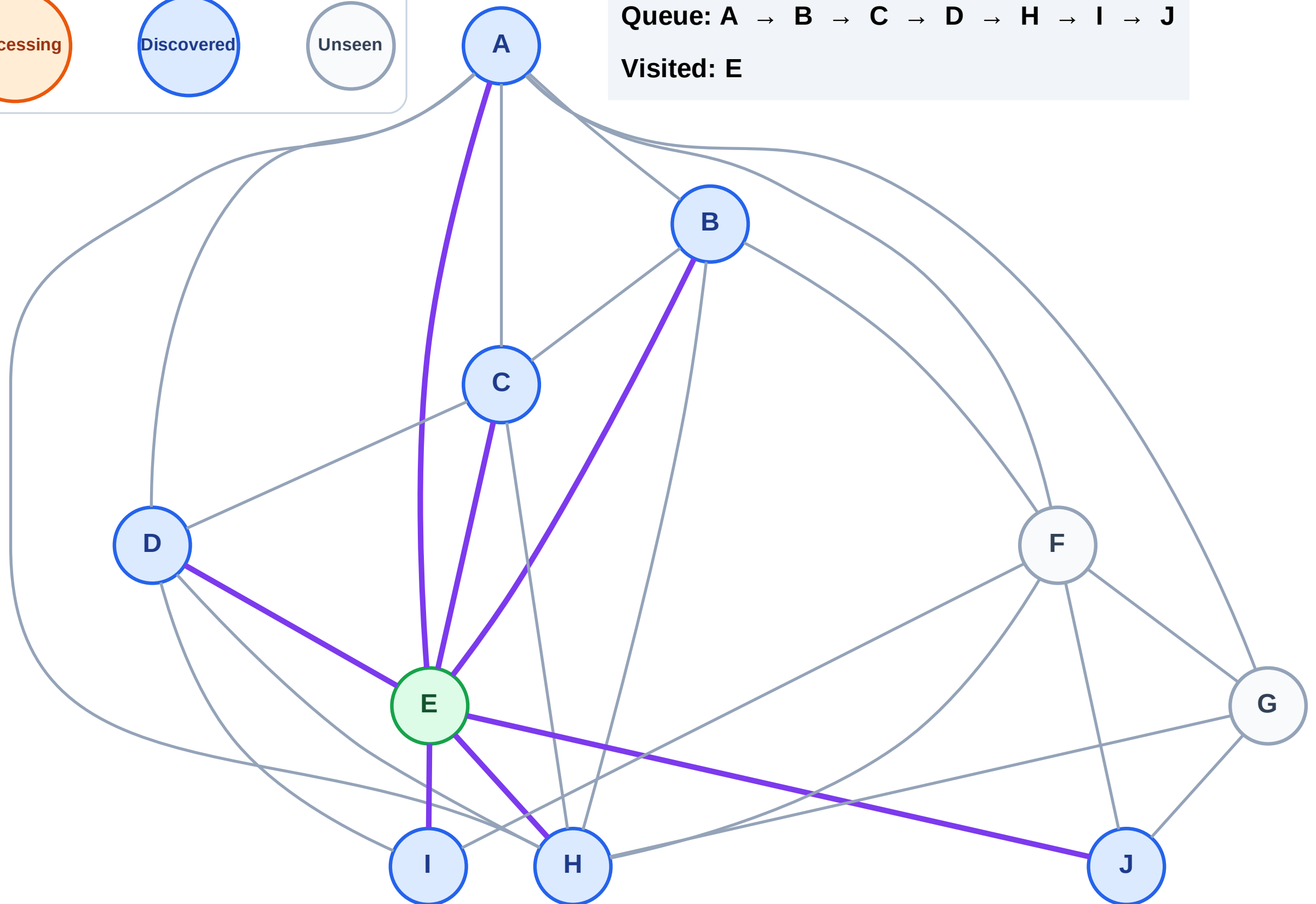
Discovered



Unseen

Queue: A → B → C → D → H → I → J

Visited: E



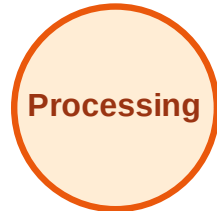
BFS Traversal

Step 4: Process node A

Legend



Complete



Processing



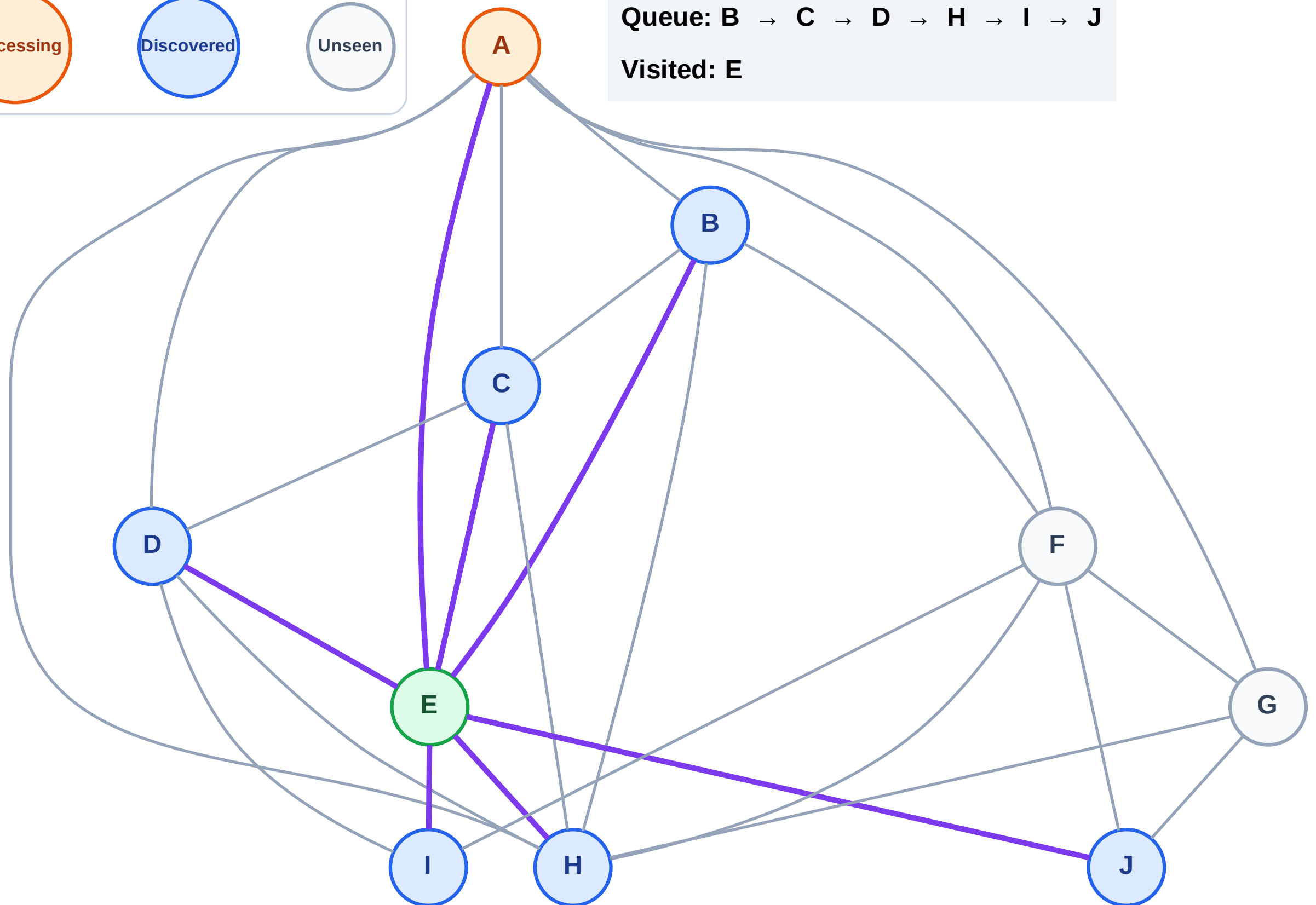
Discovered



Unseen

Queue: B → C → D → H → I → J

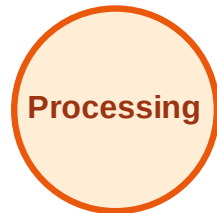
Visited: E



BFS Traversal

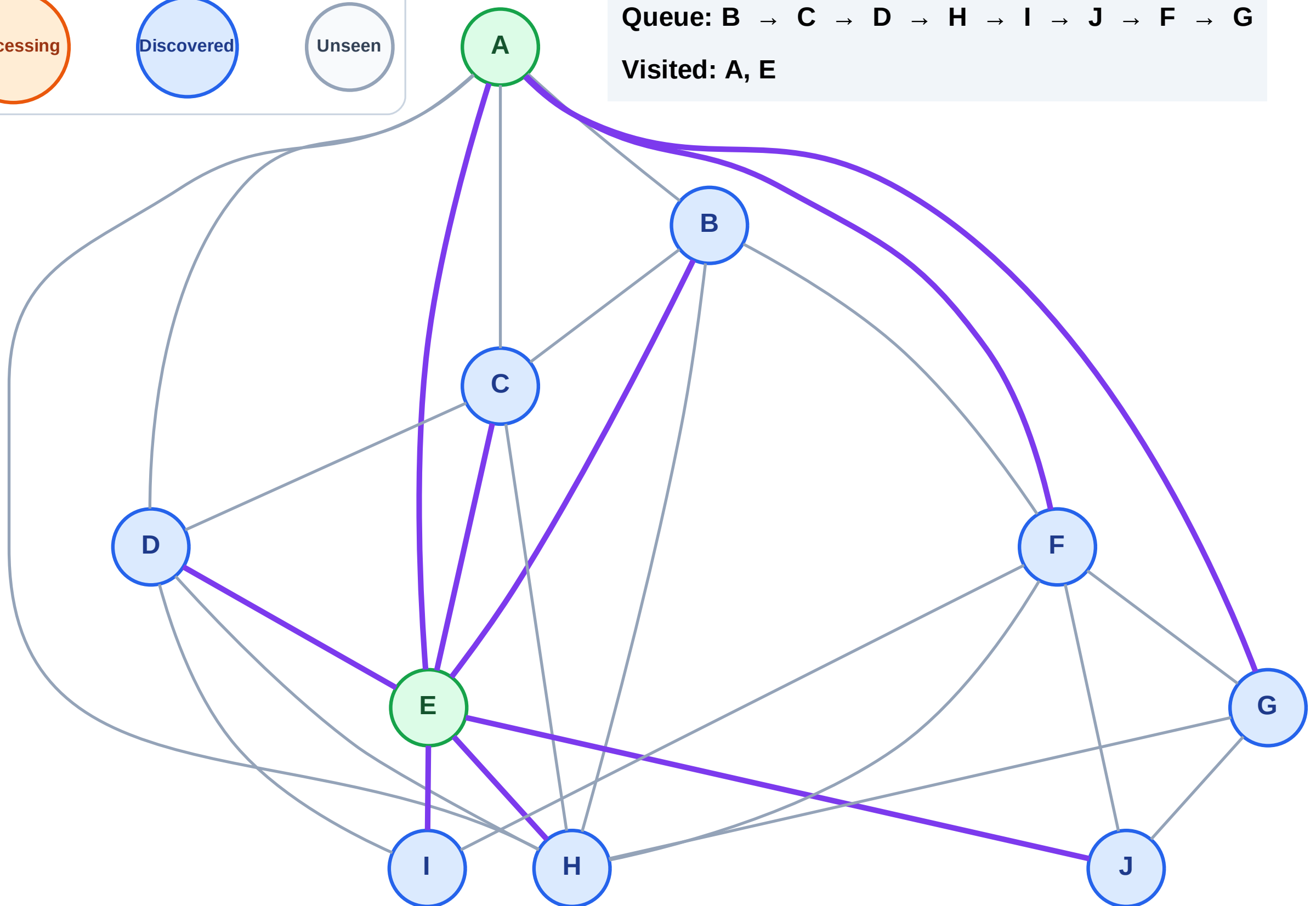
Step 5: Discover F, G from A

Legend



Queue: B → C → D → H → I → J → F → G

Visited: A, E



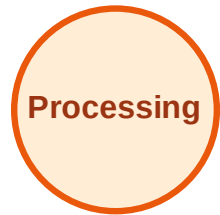
BFS Traversal

Step 6: Process node B

Legend



Complete



Processing



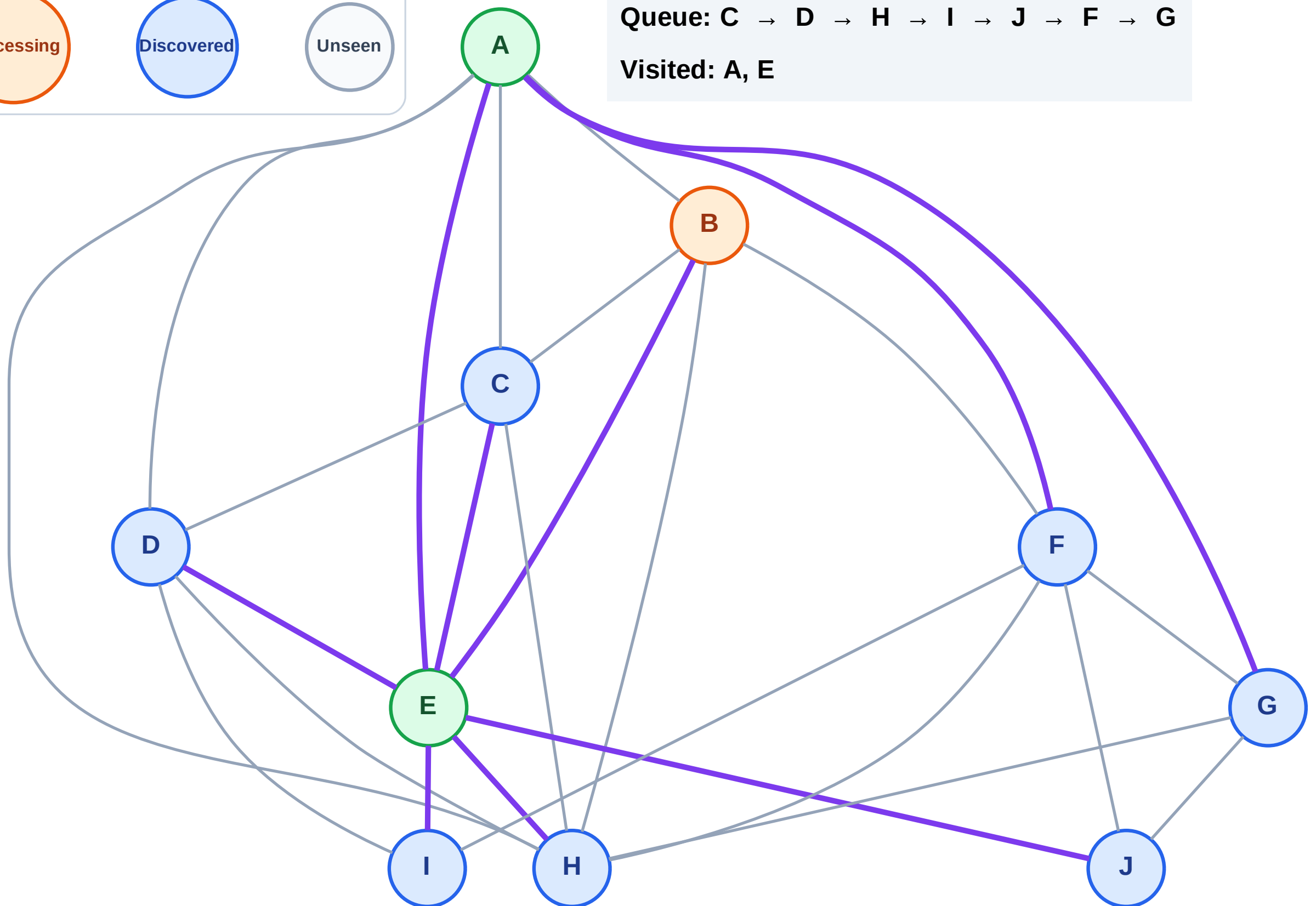
Discovered



Unseen

Queue: C → D → H → I → J → F → G

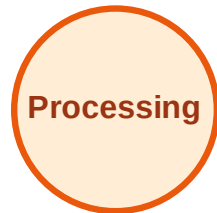
Visited: A, E



BFS Traversal

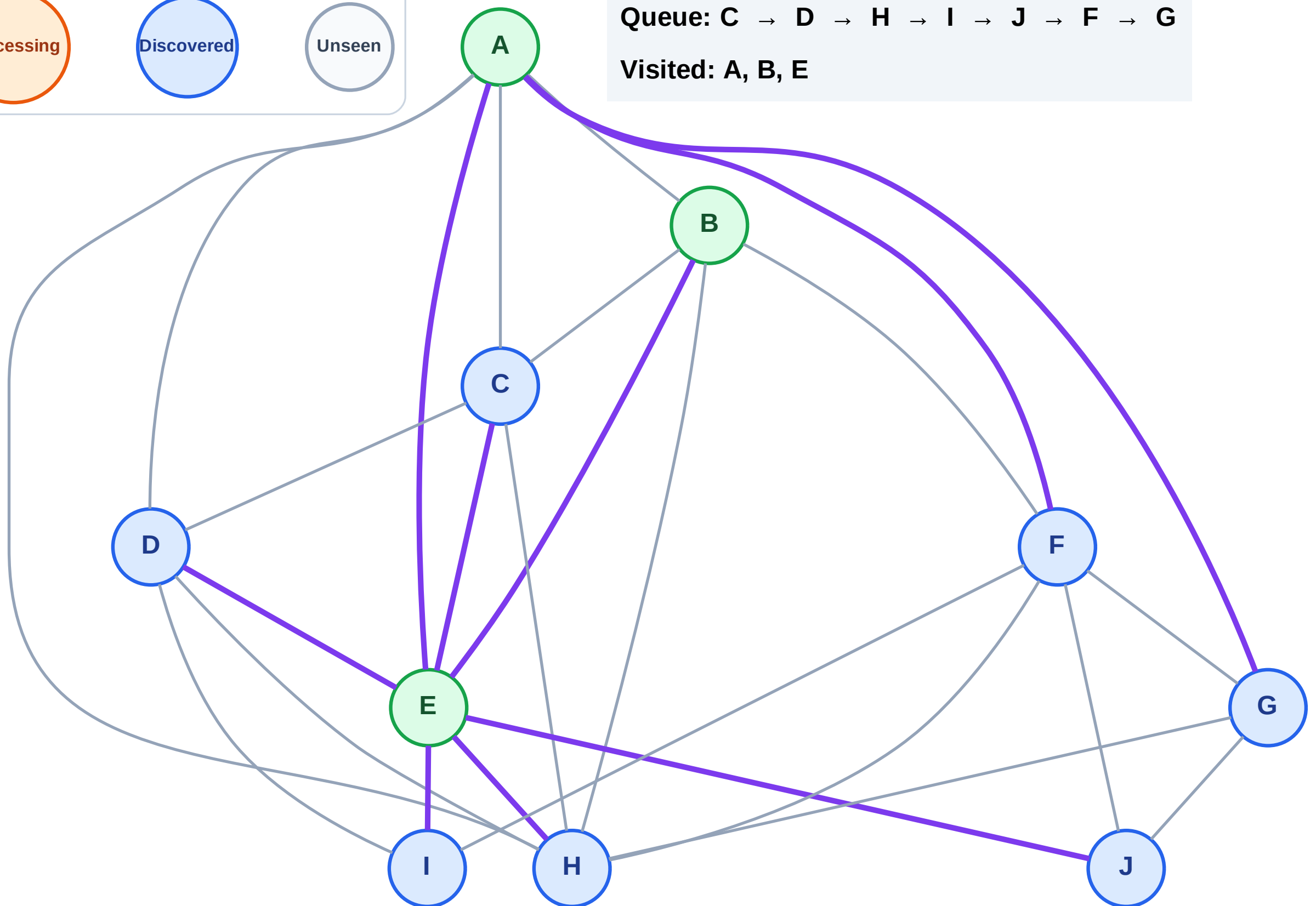
Step 7: No new neighbors from B

Legend



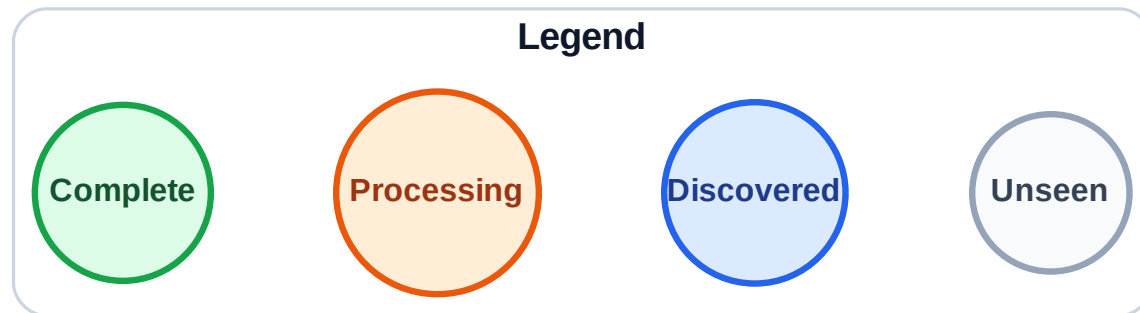
Queue: C → D → H → I → J → F → G

Visited: A, B, E



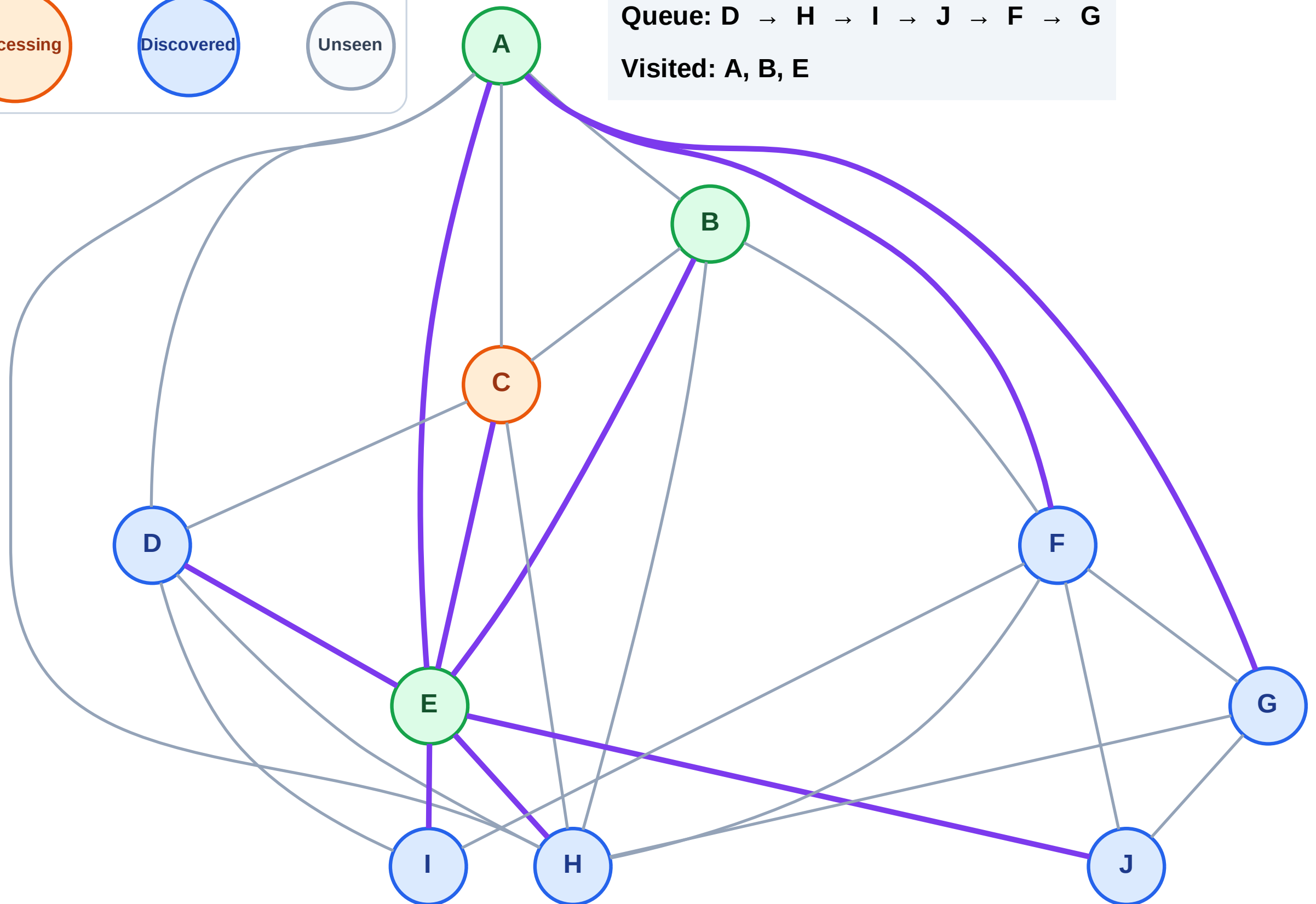
BFS Traversal

Step 8: Process node C



Queue: D → H → I → J → F → G

Visited: A, B, E



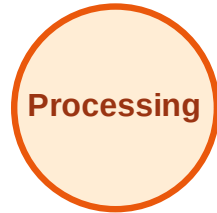
BFS Traversal

Step 9: No new neighbors from C

Legend



Complete



Processing



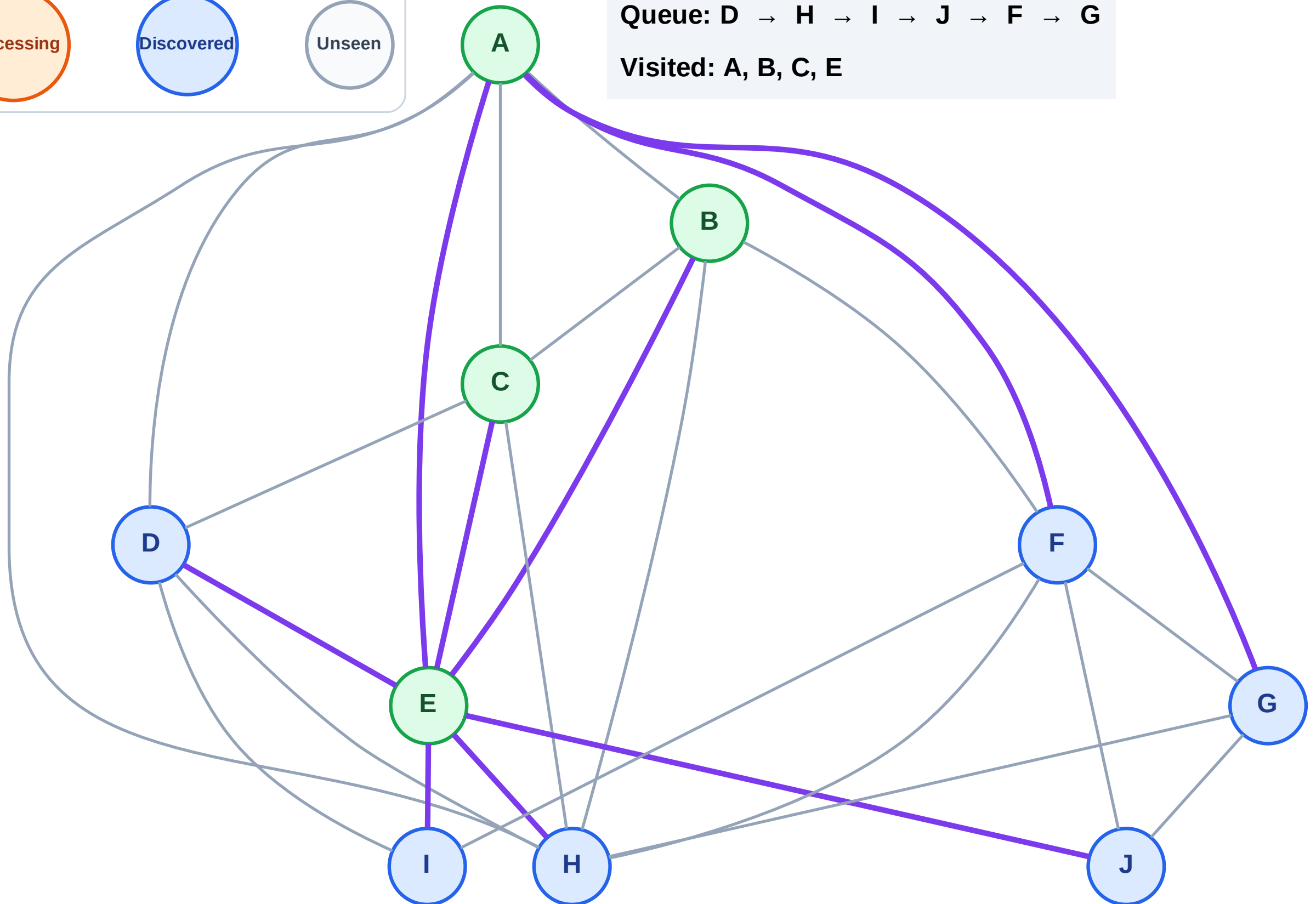
Discovered



Unseen

Queue: D → H → I → J → F → G

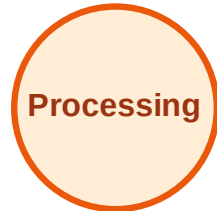
Visited: A, B, C, E



BFS Traversal

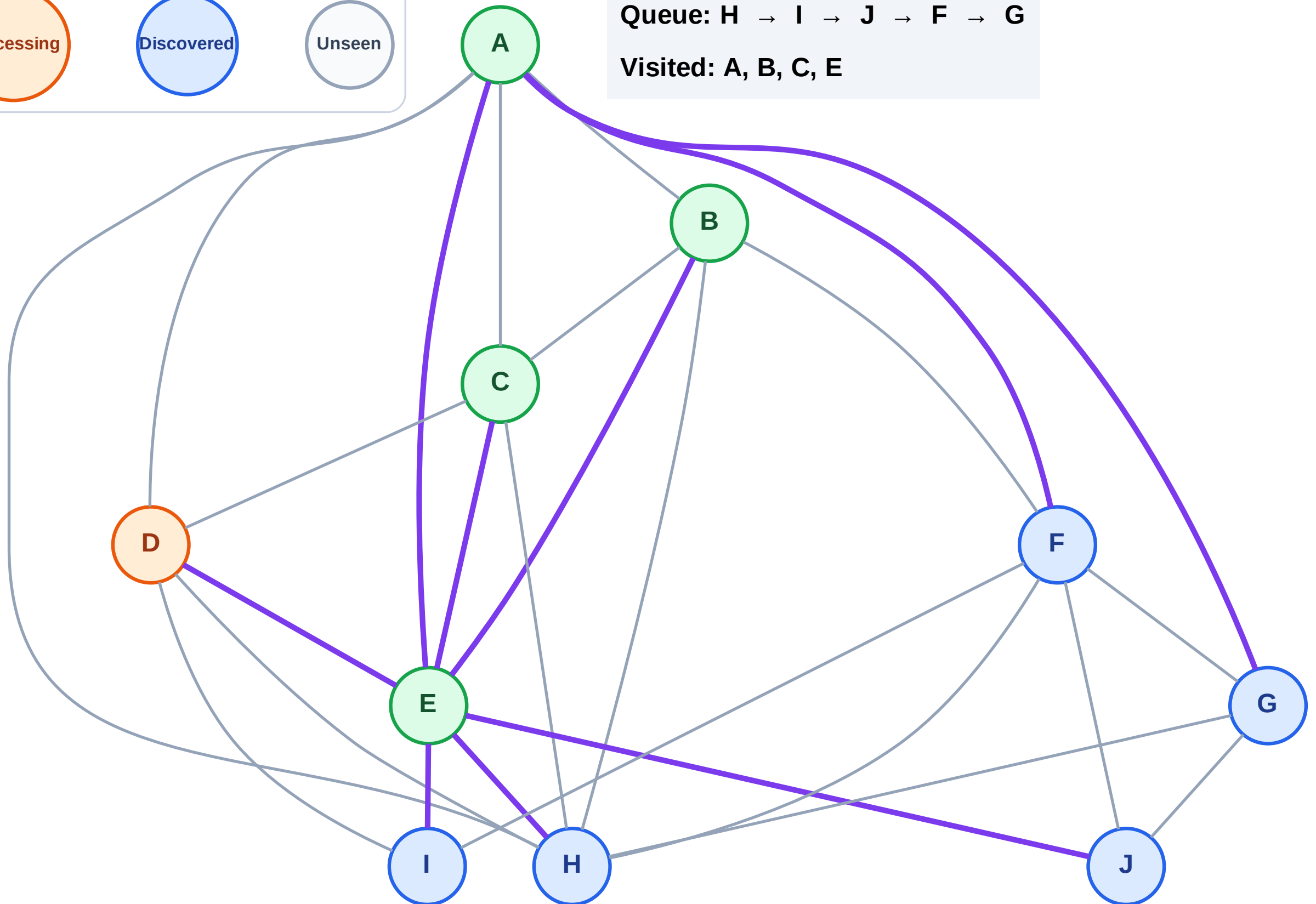
Step 10: Process node D

Legend



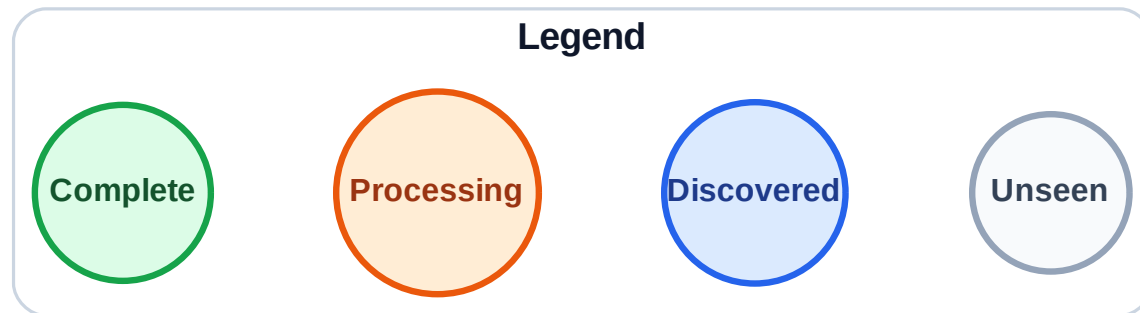
Queue: H → I → J → F → G

Visited: A, B, C, E



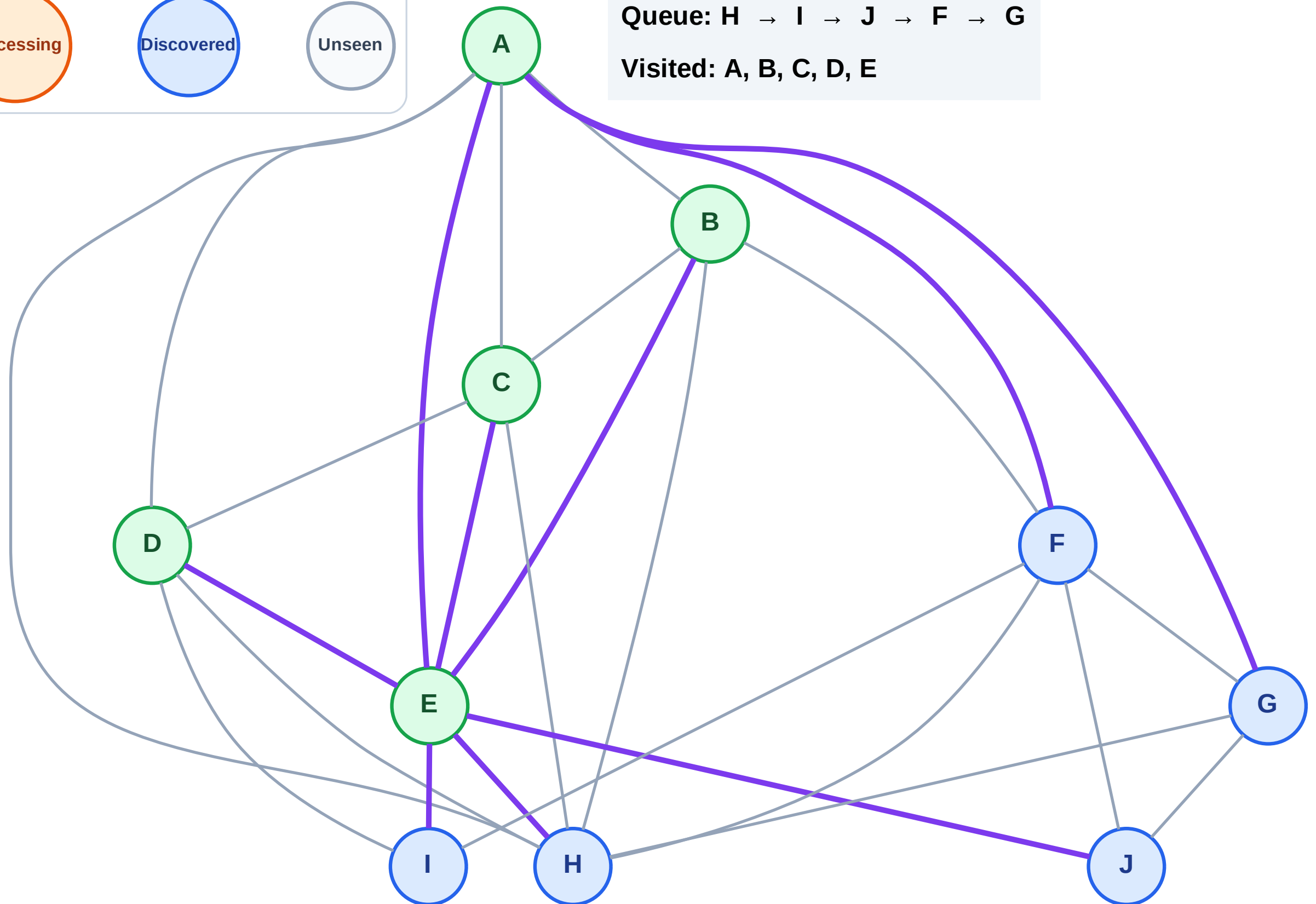
BFS Traversal

Step 11: No new neighbors from D



Queue: H → I → J → F → G

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node H

Legend

Complete

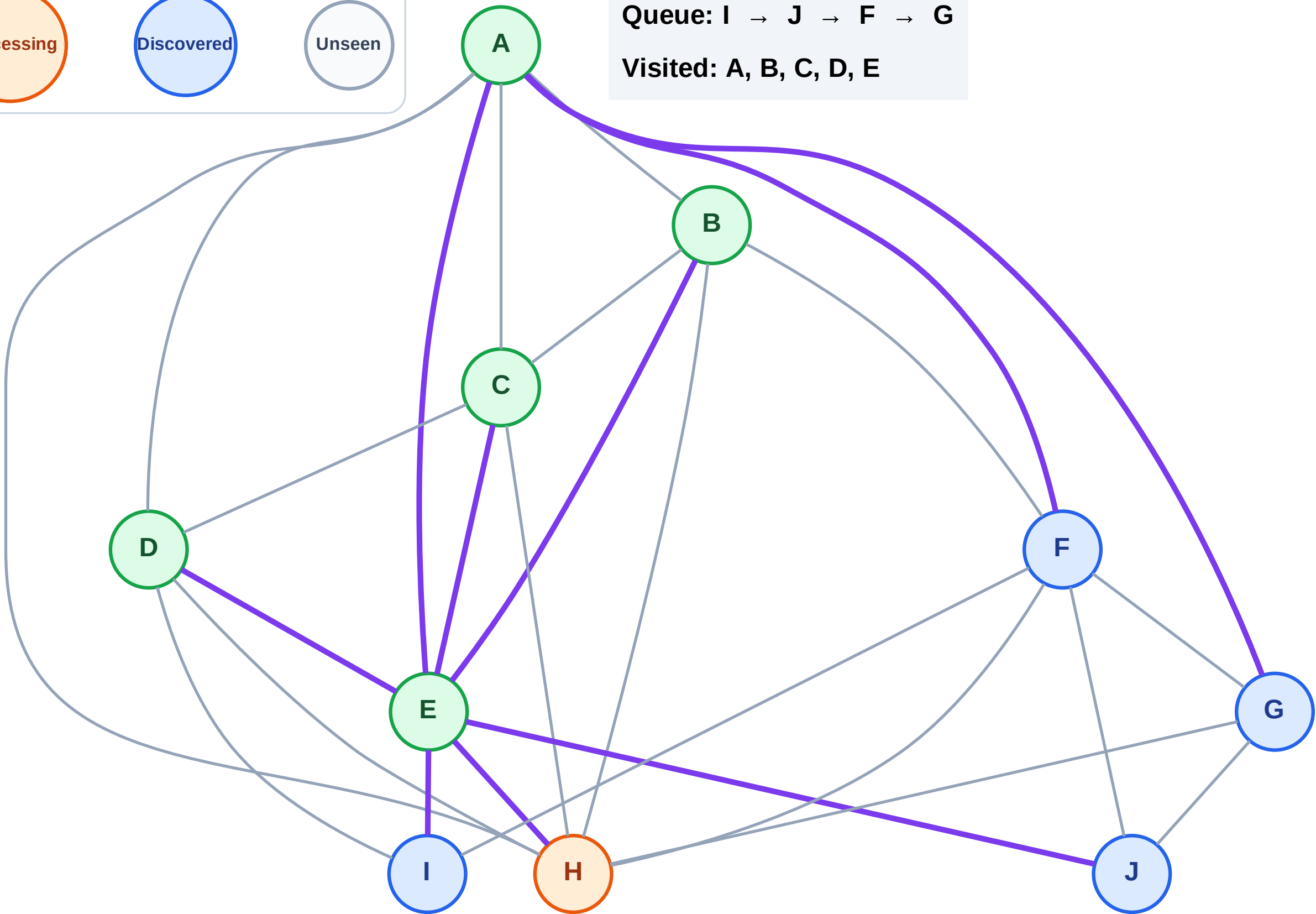
Processing

Discovered

Unseen

Queue: I → J → F → G

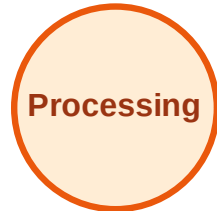
Visited: A, B, C, D, E



BFS Traversal

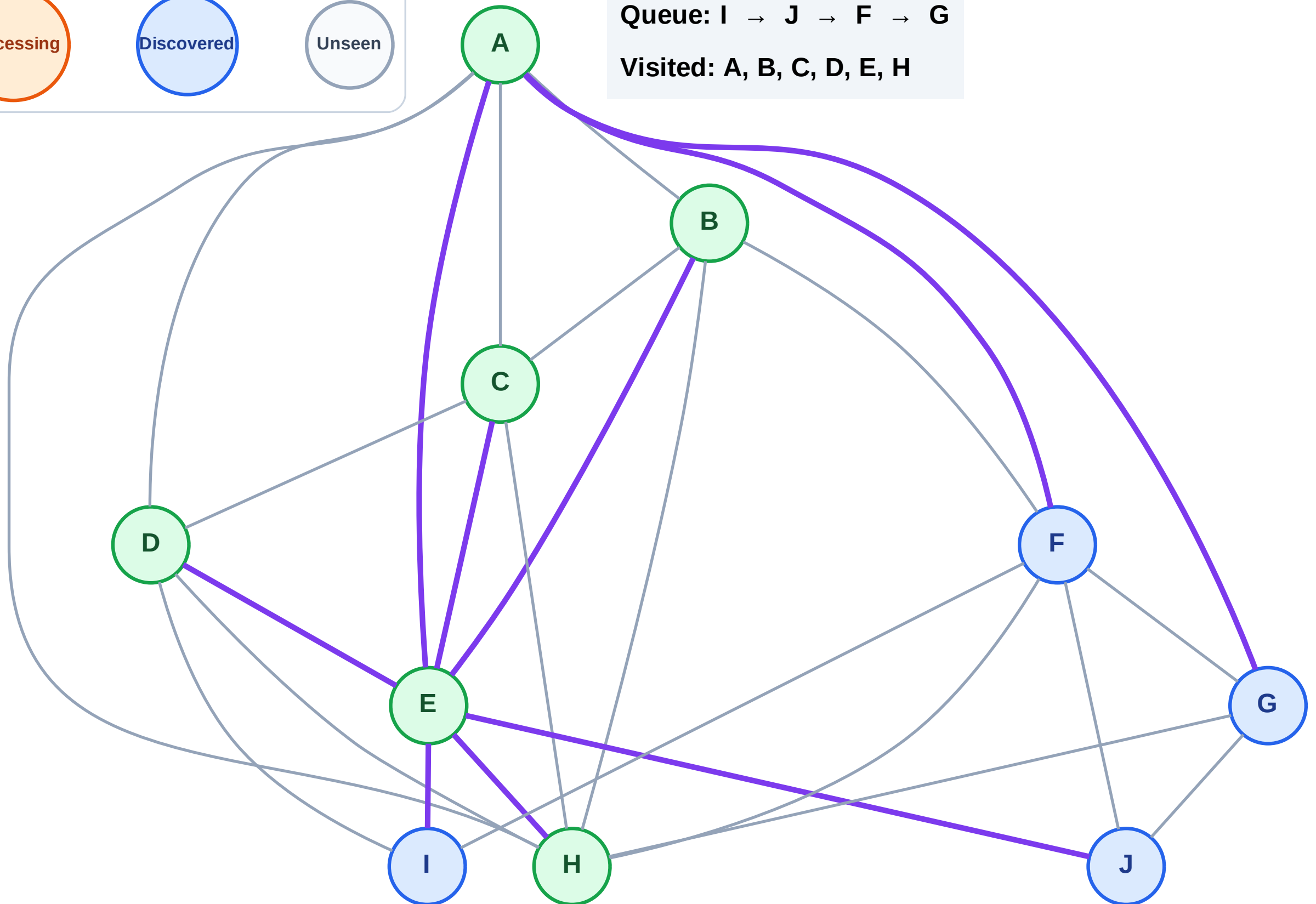
Step 13: No new neighbors from H

Legend



Queue: I → J → F → G

Visited: A, B, C, D, E, H



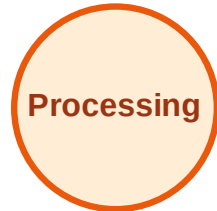
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing



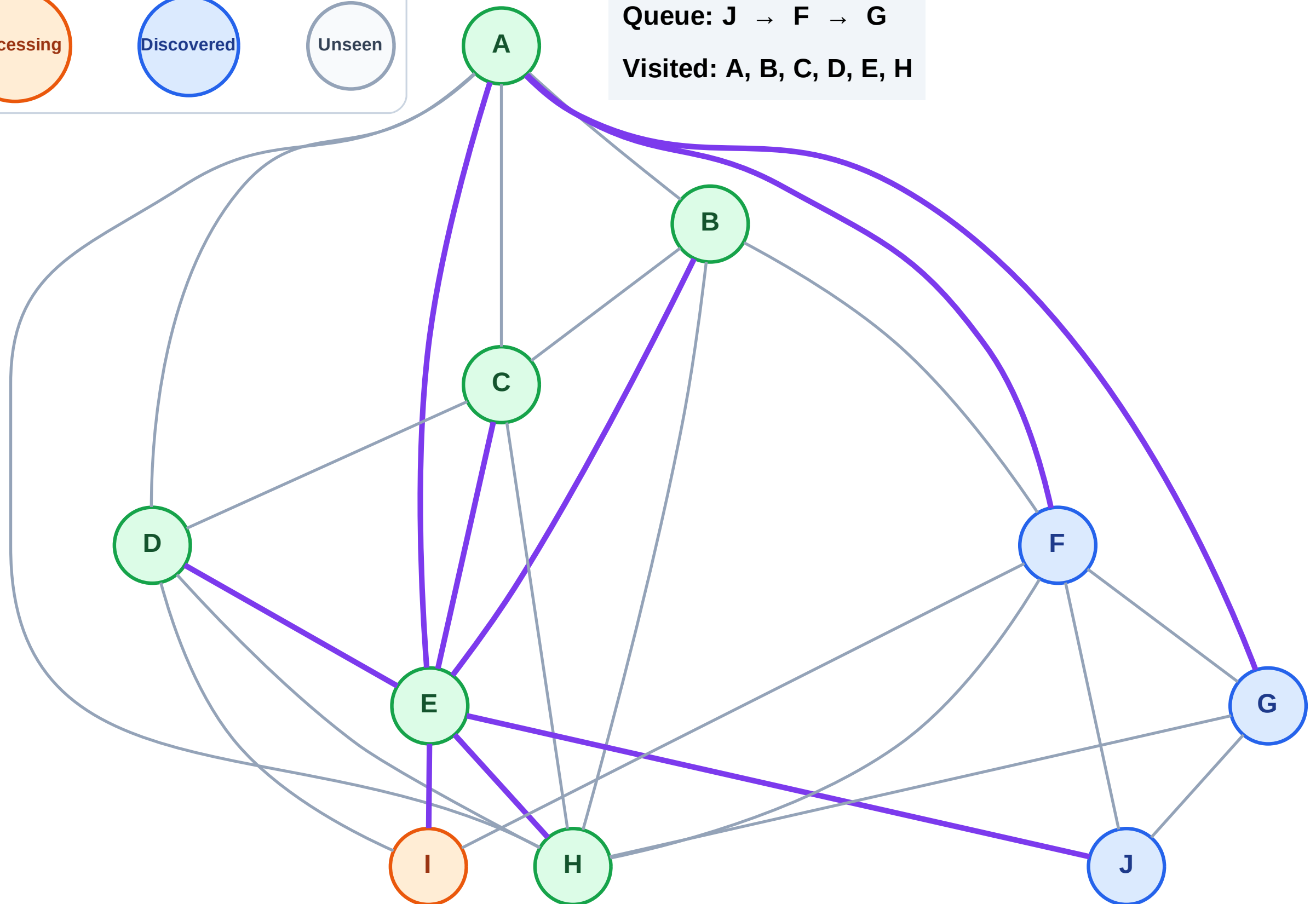
Discovered



Unseen

Queue: J → F → G

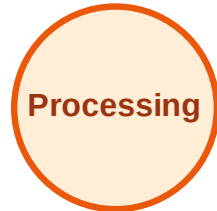
Visited: A, B, C, D, E, H



BFS Traversal

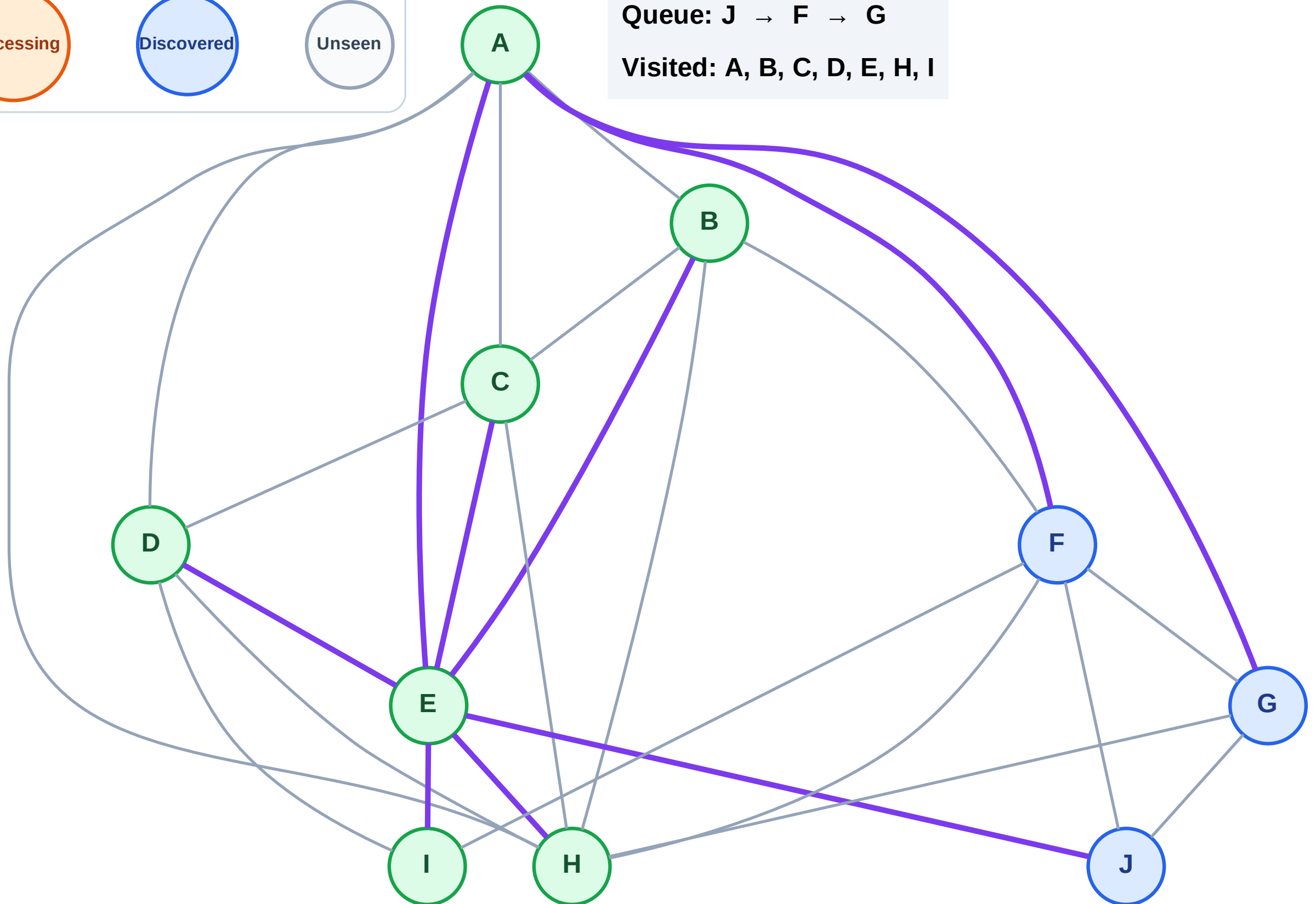
Step 15: No new neighbors from I

Legend



Queue: J → F → G

Visited: A, B, C, D, E, H, I



BFS Traversal

Step 16: Process node J

Legend

Complete

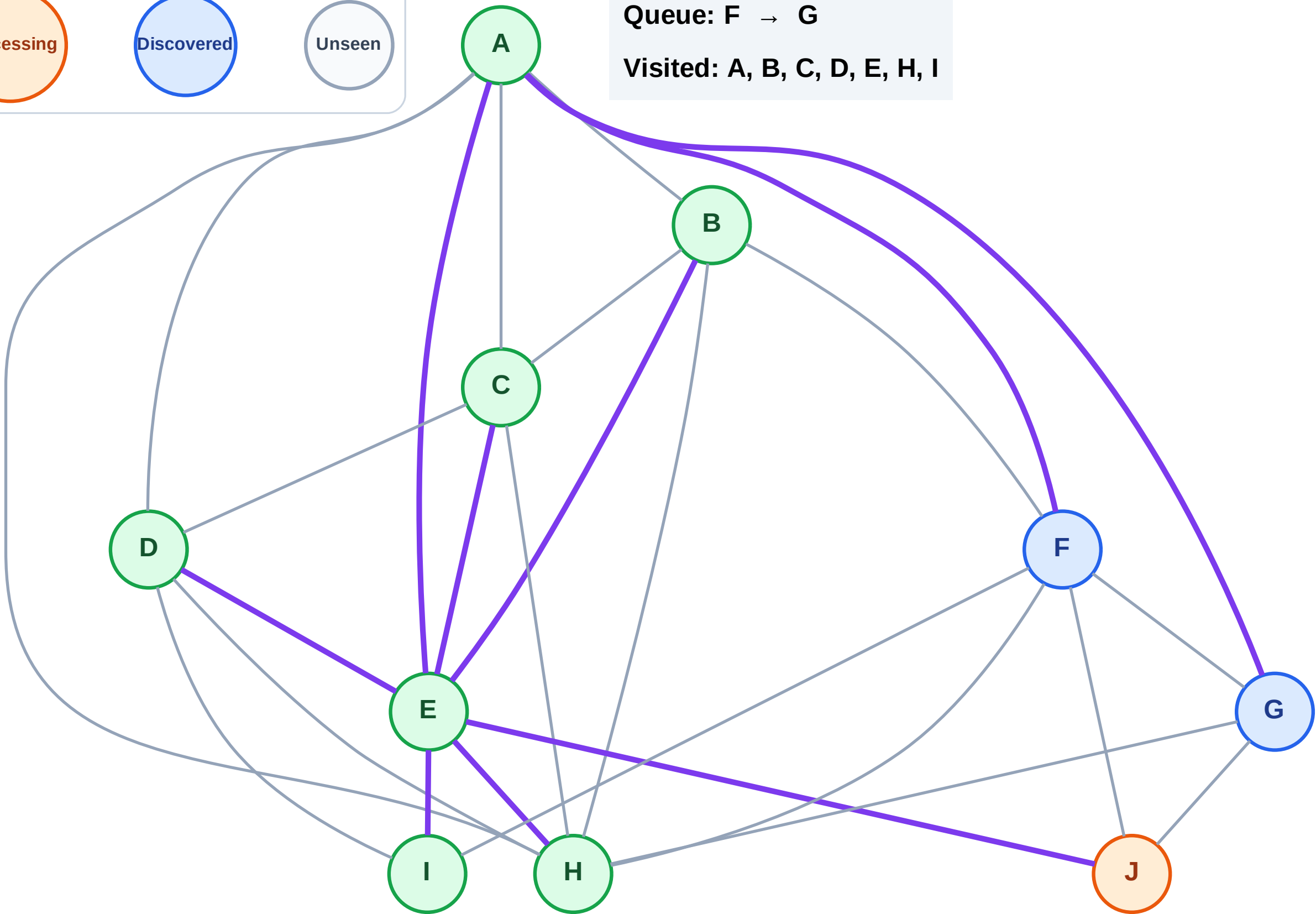
Processing

Discovered

Unseen

Queue: F → G

Visited: A, B, C, D, E, H, I



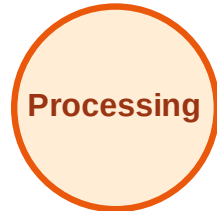
BFS Traversal

Step 17: No new neighbors from J

Legend



Complete



Processing



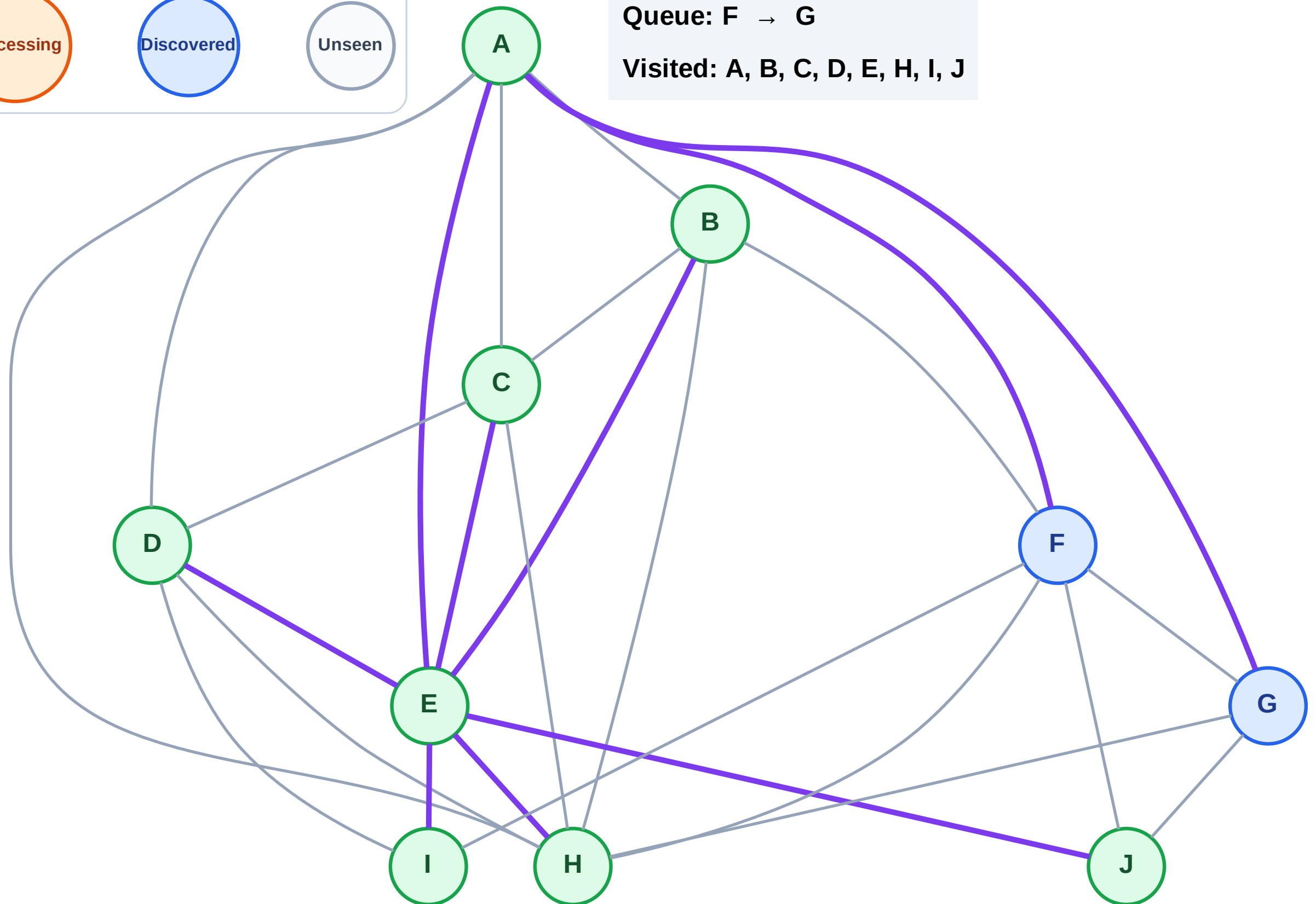
Discovered



Unseen

Queue: F → G

Visited: A, B, C, D, E, H, I, J



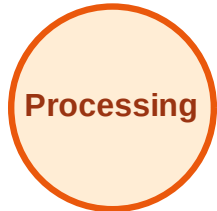
BFS Traversal

Step 18: Process node F

Legend



Complete



Processing



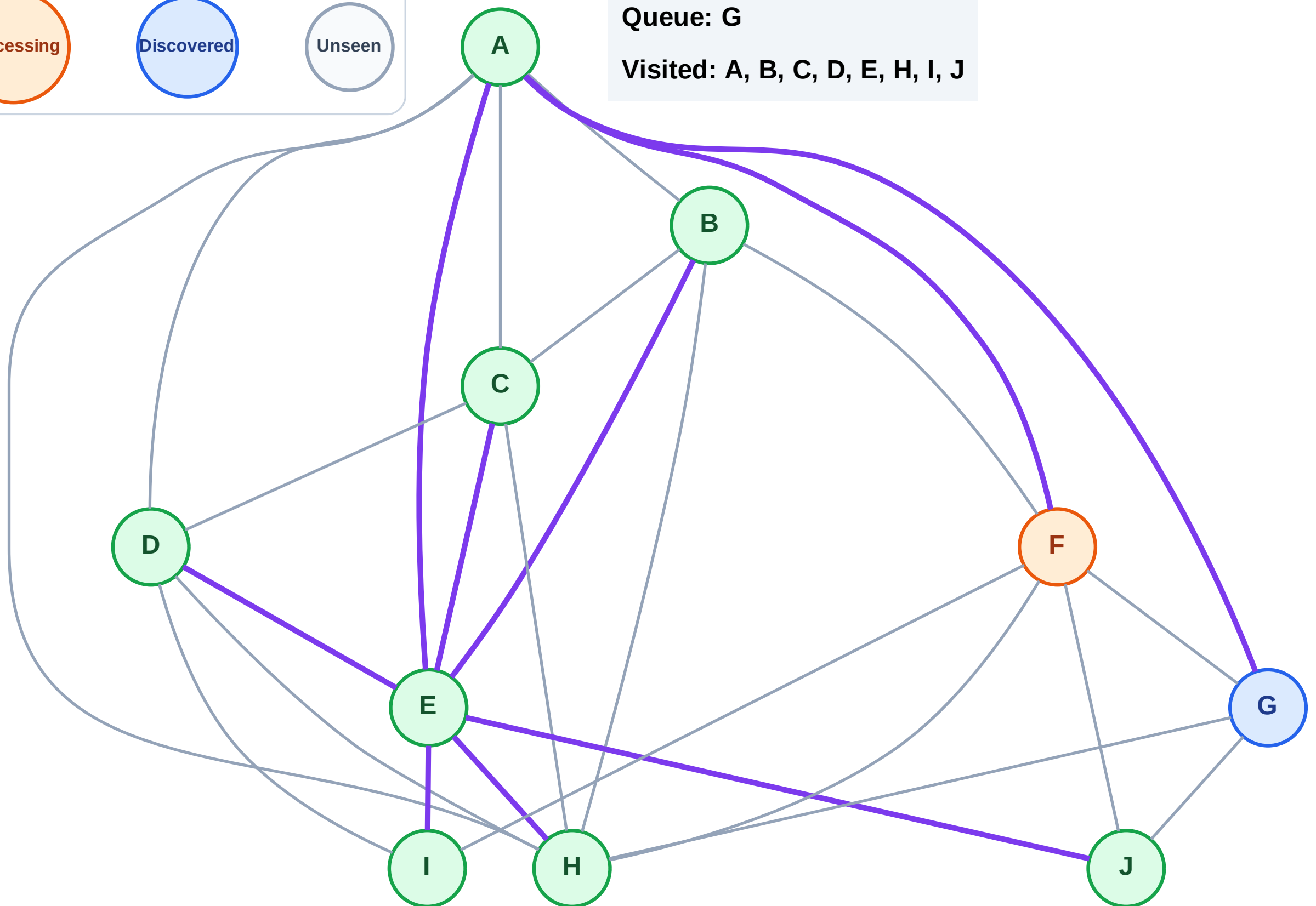
Discovered



Unseen

Queue: G

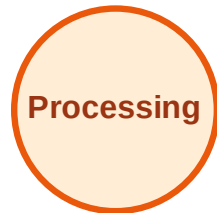
Visited: A, B, C, D, E, H, I, J



BFS Traversal

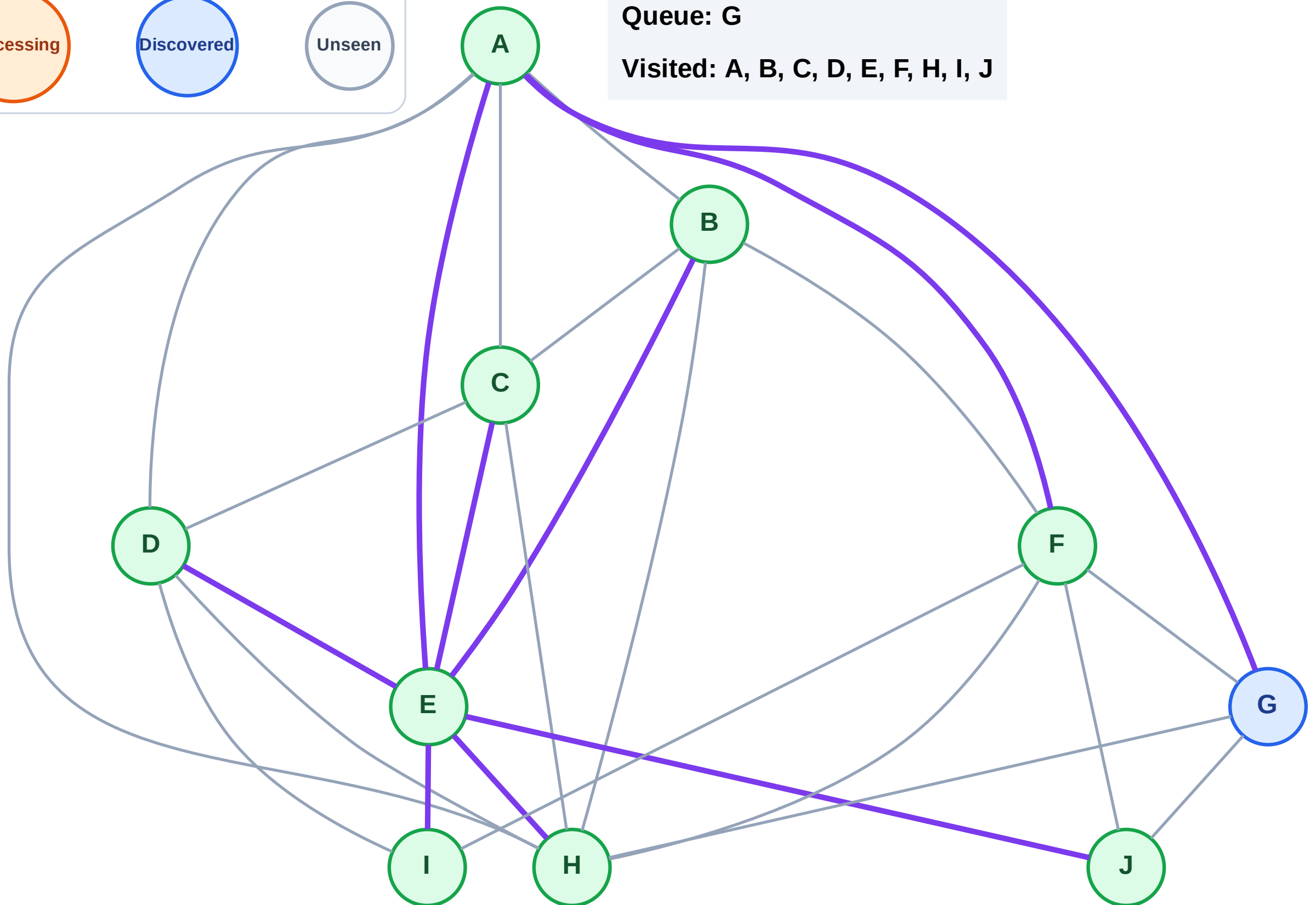
Step 19: No new neighbors from F

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

Step 20: Process node G

Legend

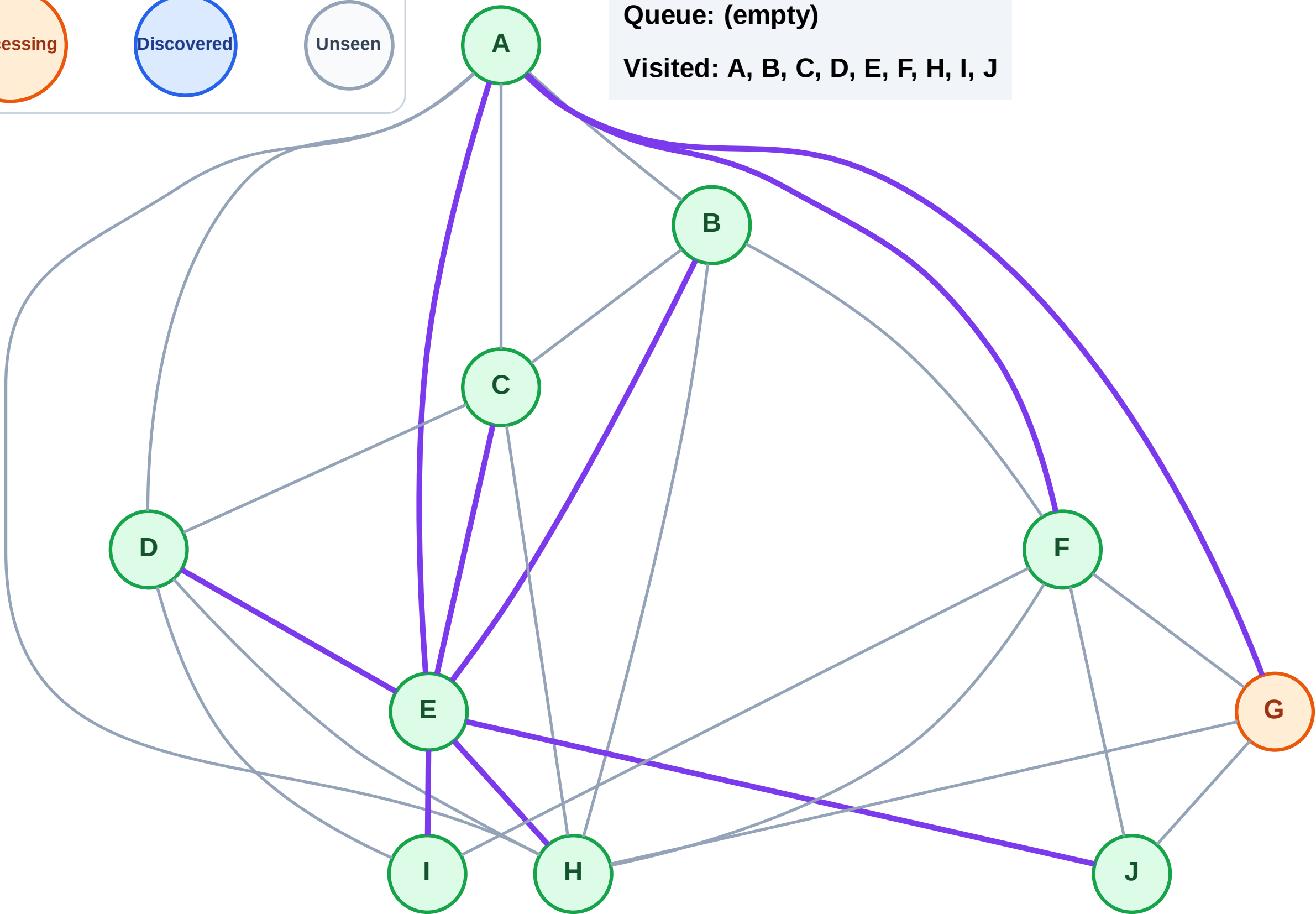
Complete

Processing

Discovered

Unseen

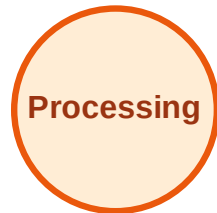
Queue: (empty)
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

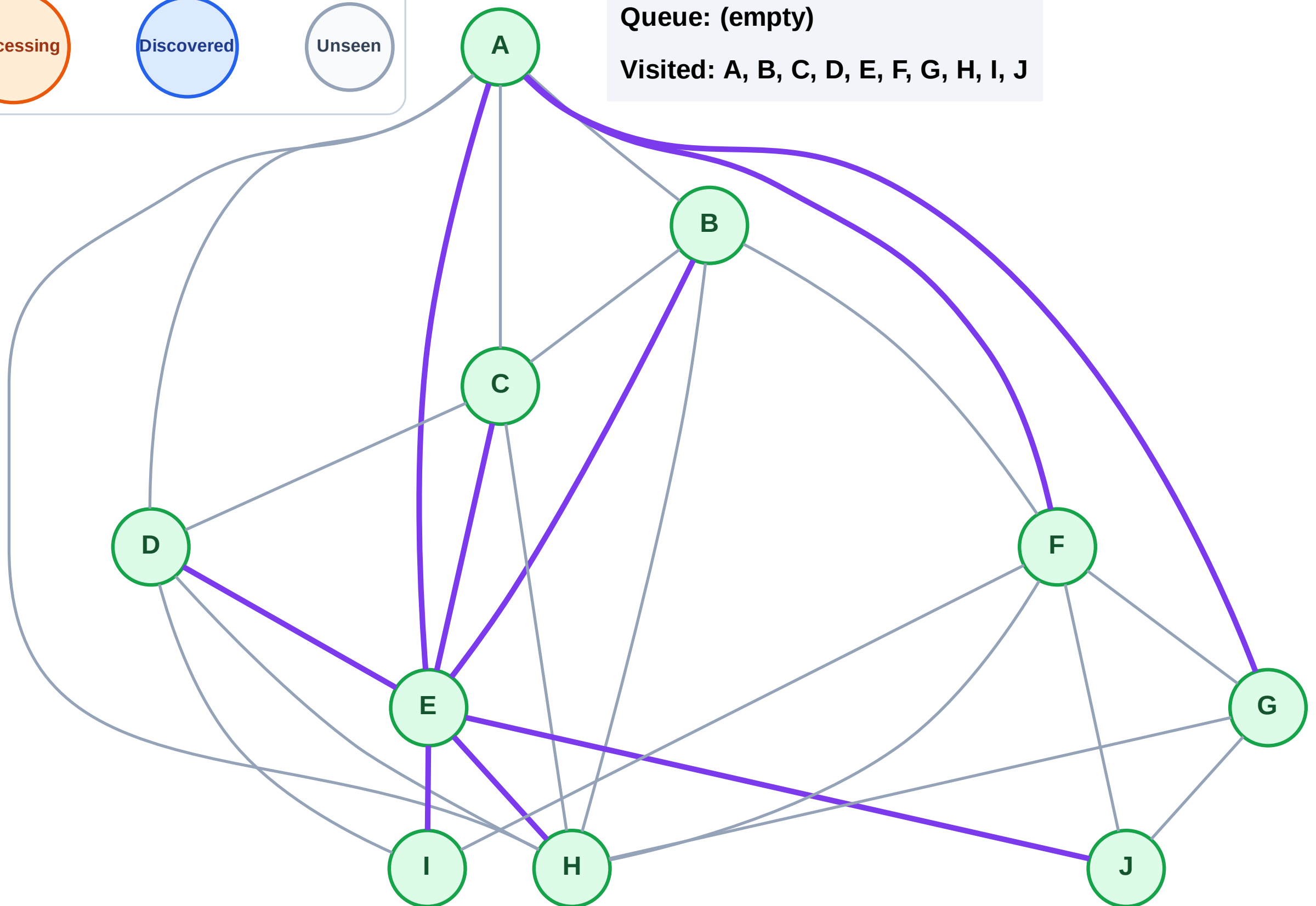
Step 21: No new neighbors from G

Legend



Queue: (empty)

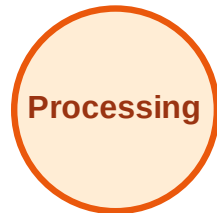
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

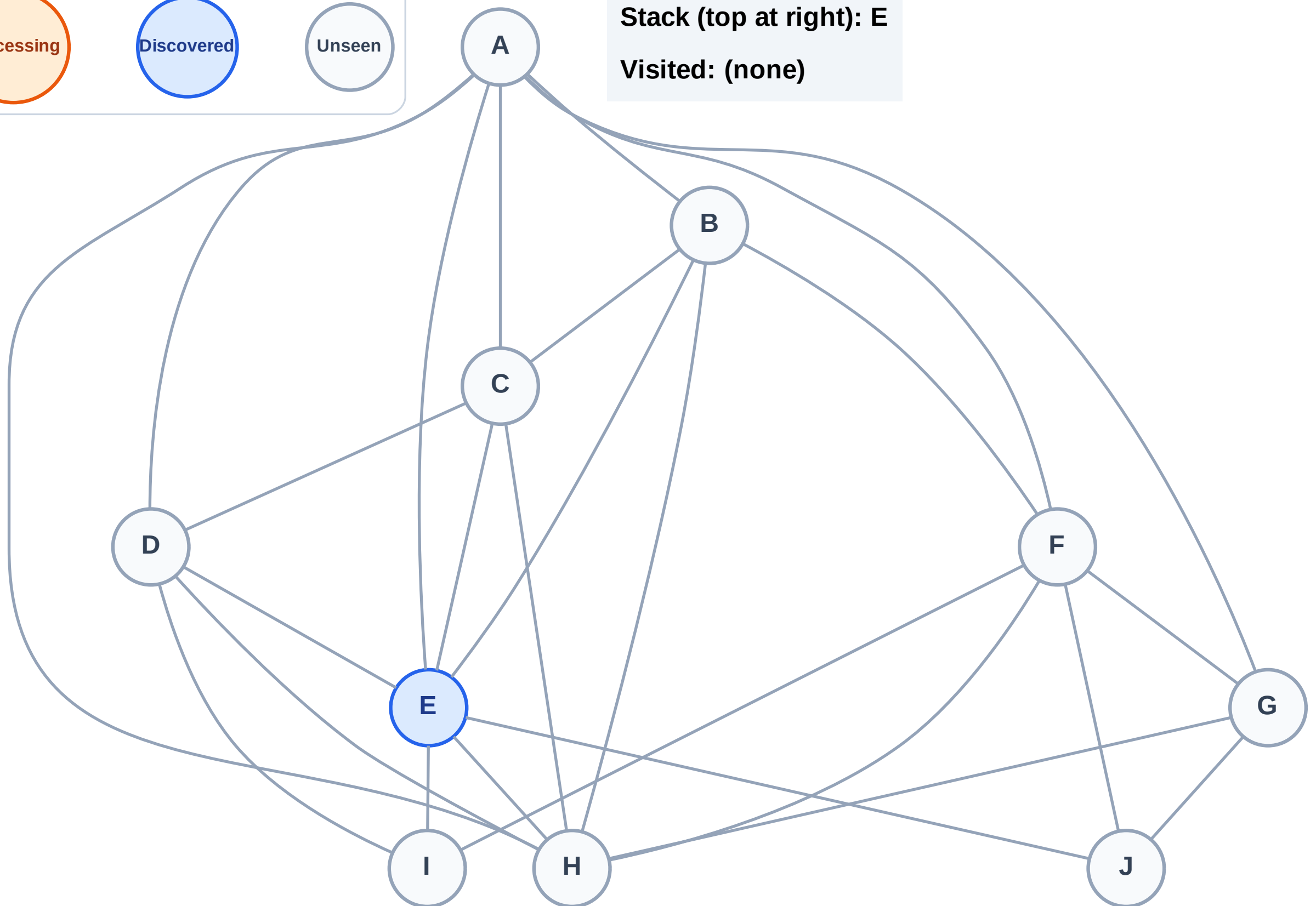
Step 1: Start at E

Legend



Stack (top at right): E

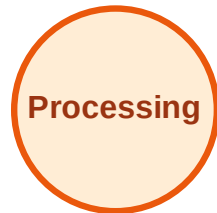
Visited: (none)



DFS Traversal

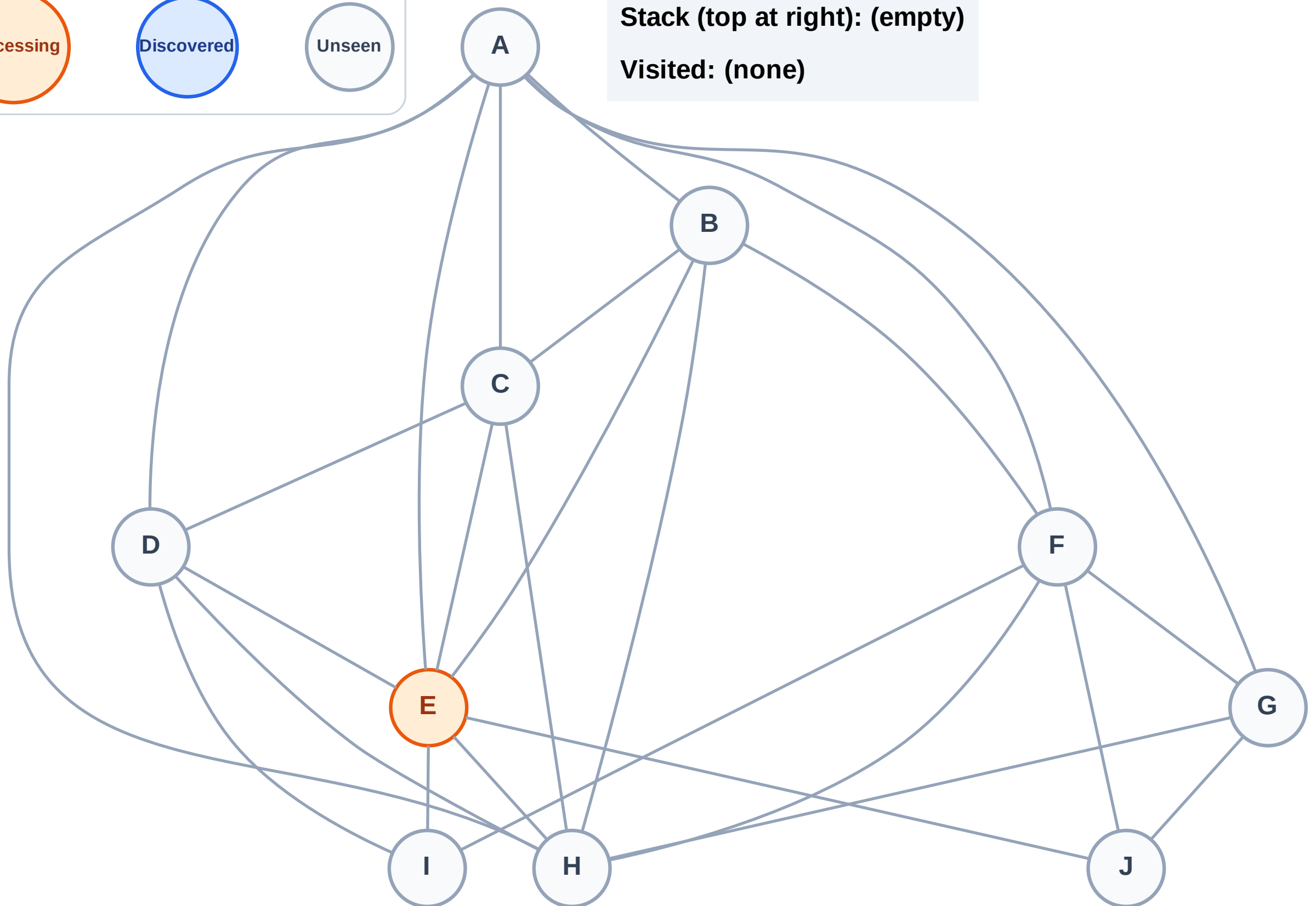
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



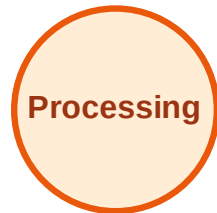
DFS Traversal

Step 3: Discover J, I, H, D, C, B, A from E

Legend



Complete



Processing



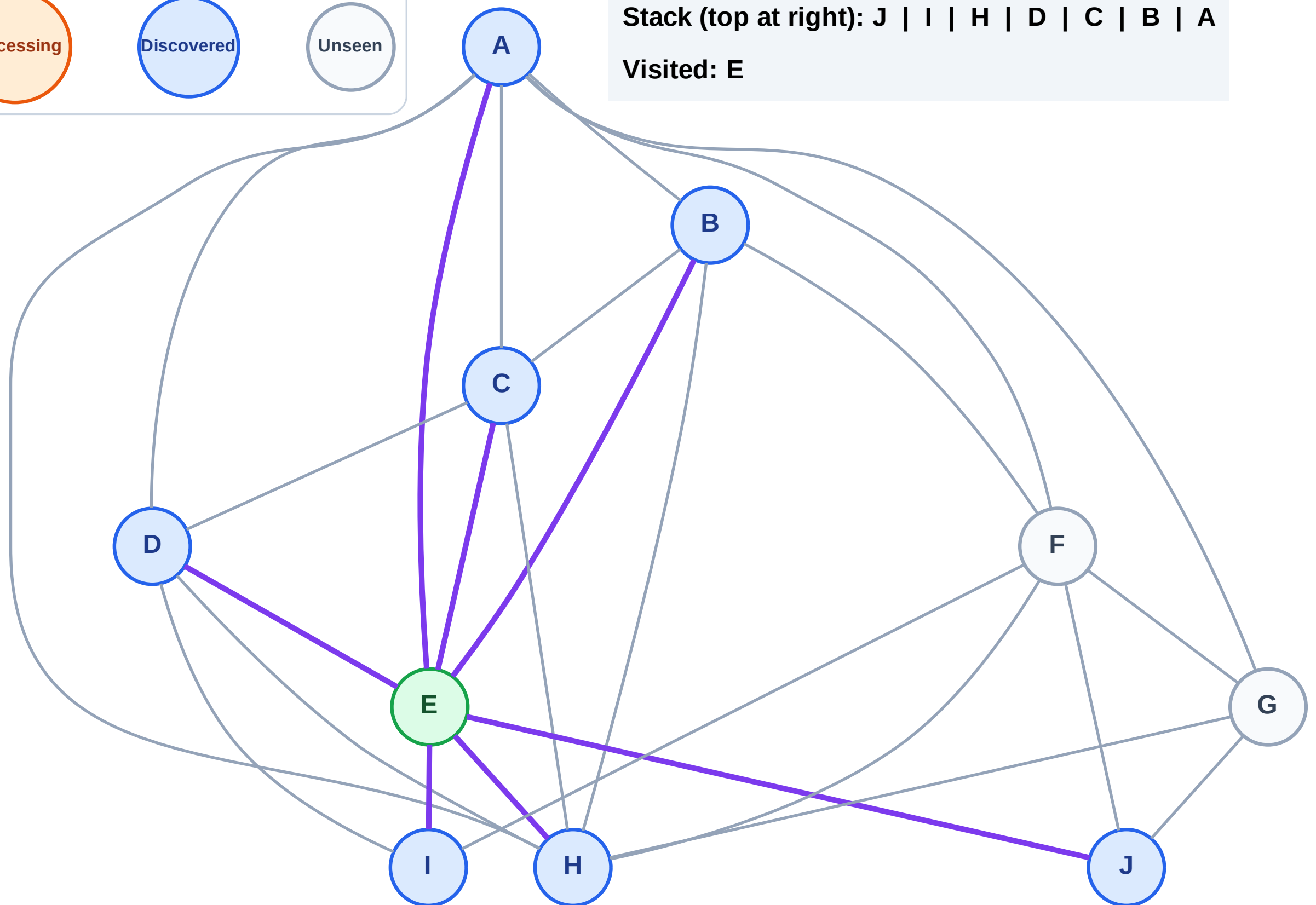
Discovered



Unseen

Stack (top at right): J | I | H | D | C | B | A

Visited: E



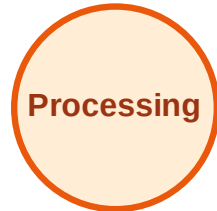
DFS Traversal

Step 4: Process node A

Legend



Complete



Processing



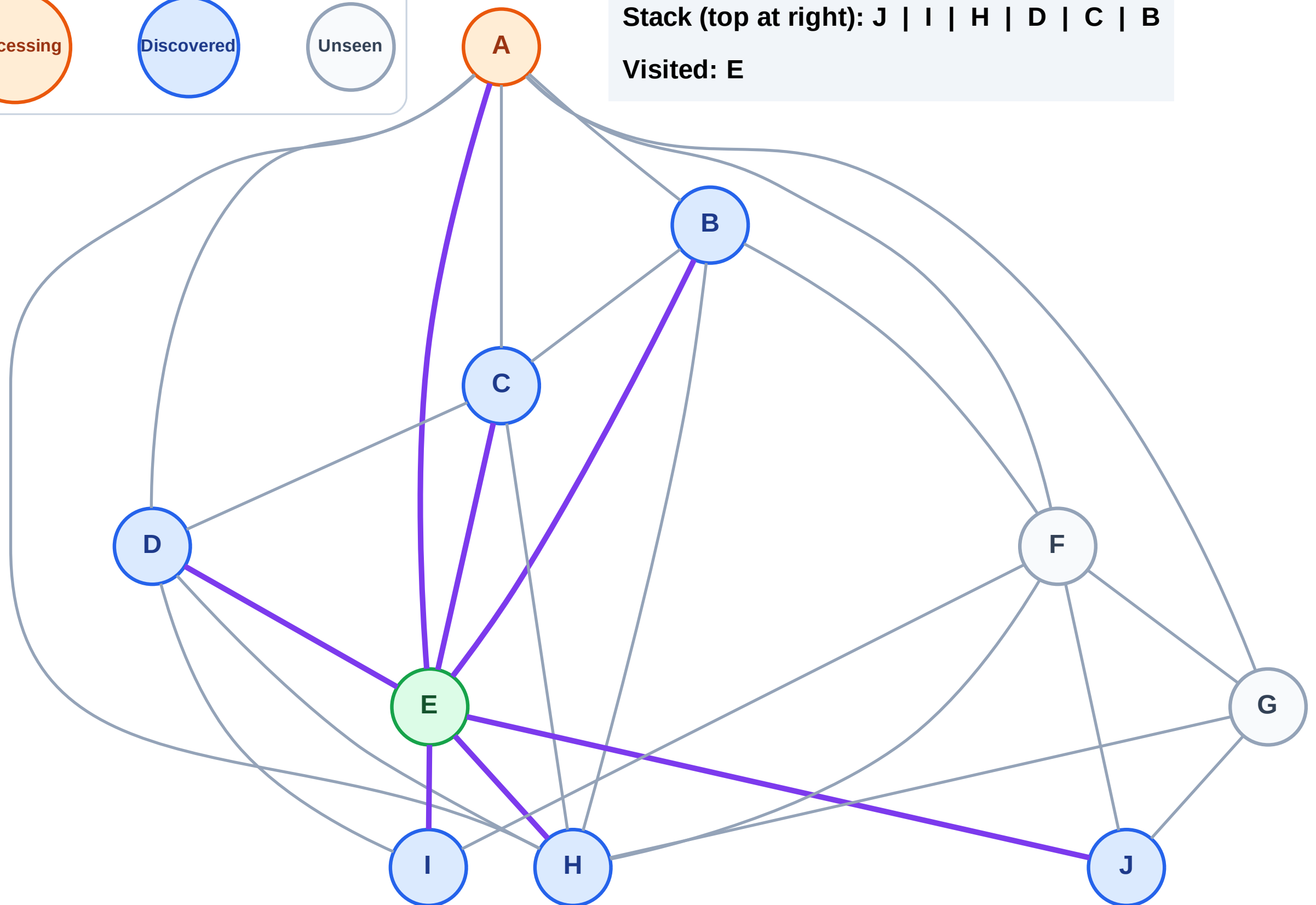
Discovered



Unseen

Stack (top at right): J | I | H | D | C | B

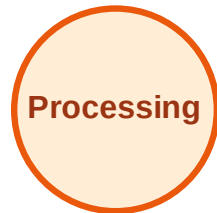
Visited: E



DFS Traversal

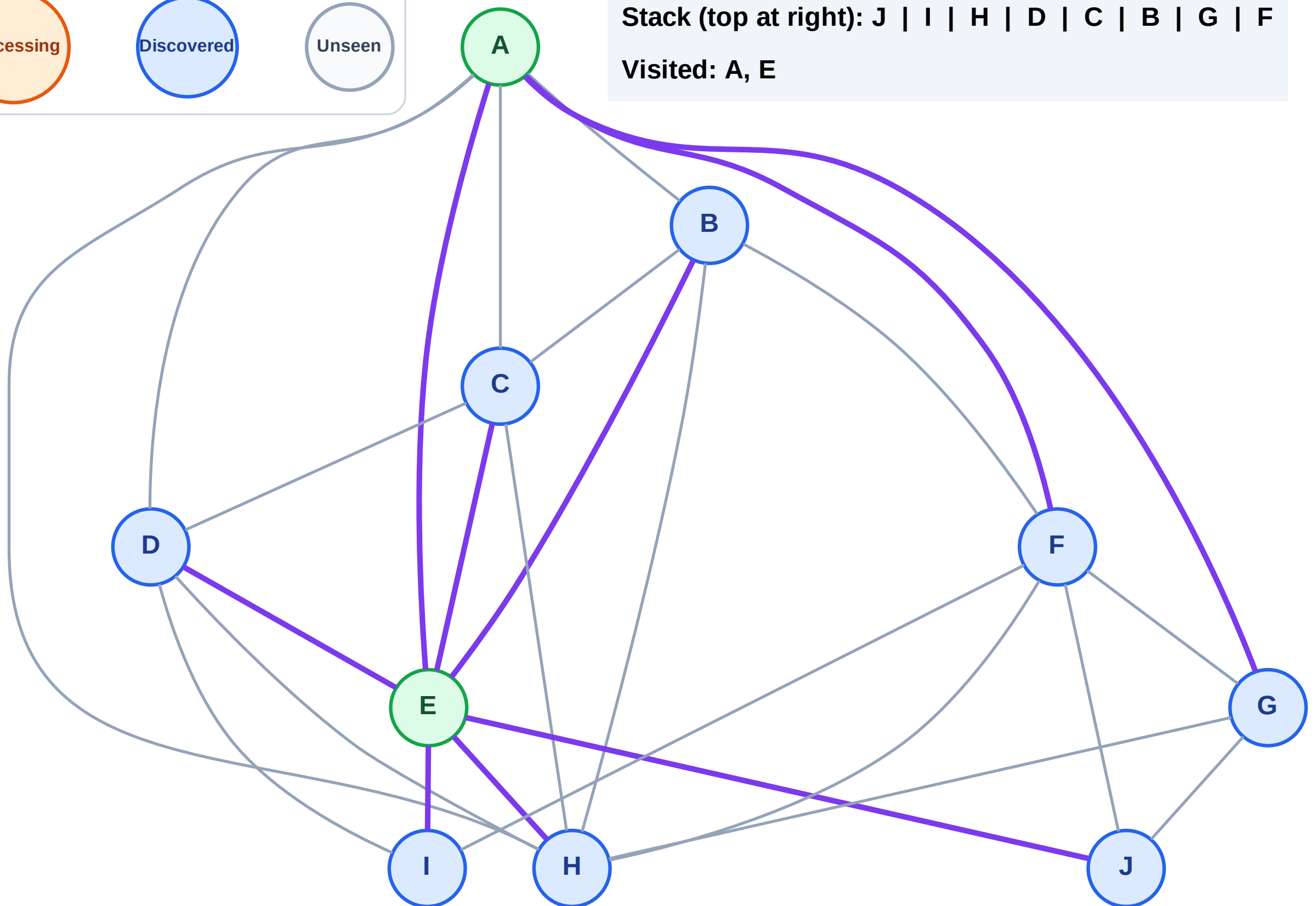
Step 5: Discover G, F from A

Legend



Stack (top at right): J | I | H | D | C | B | G | F

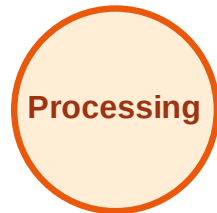
Visited: A, E



DFS Traversal

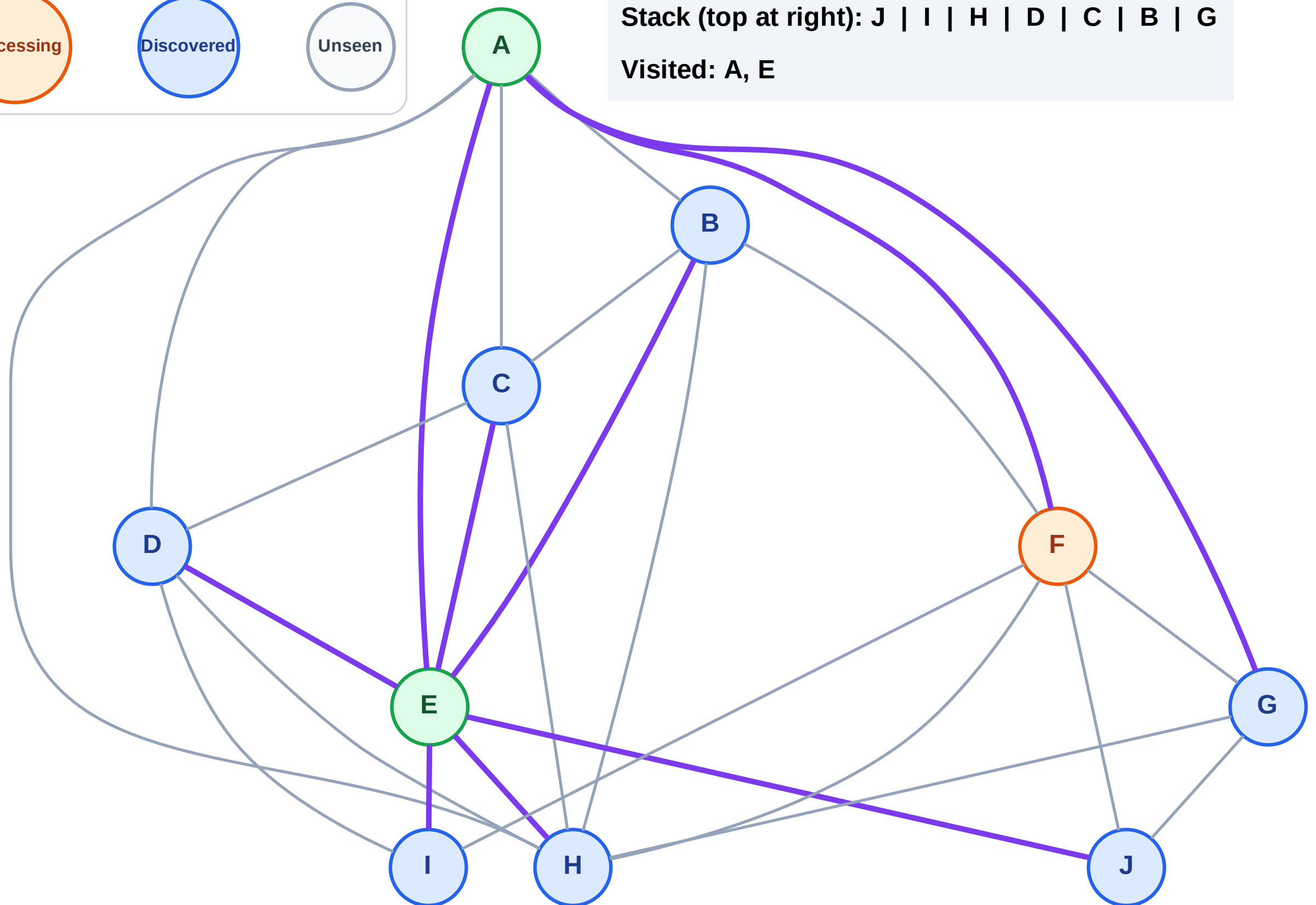
Step 6: Process node F

Legend



Stack (top at right): J | I | H | D | C | B | G

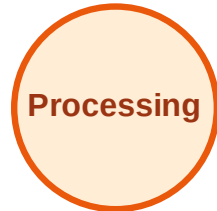
Visited: A, E



DFS Traversal

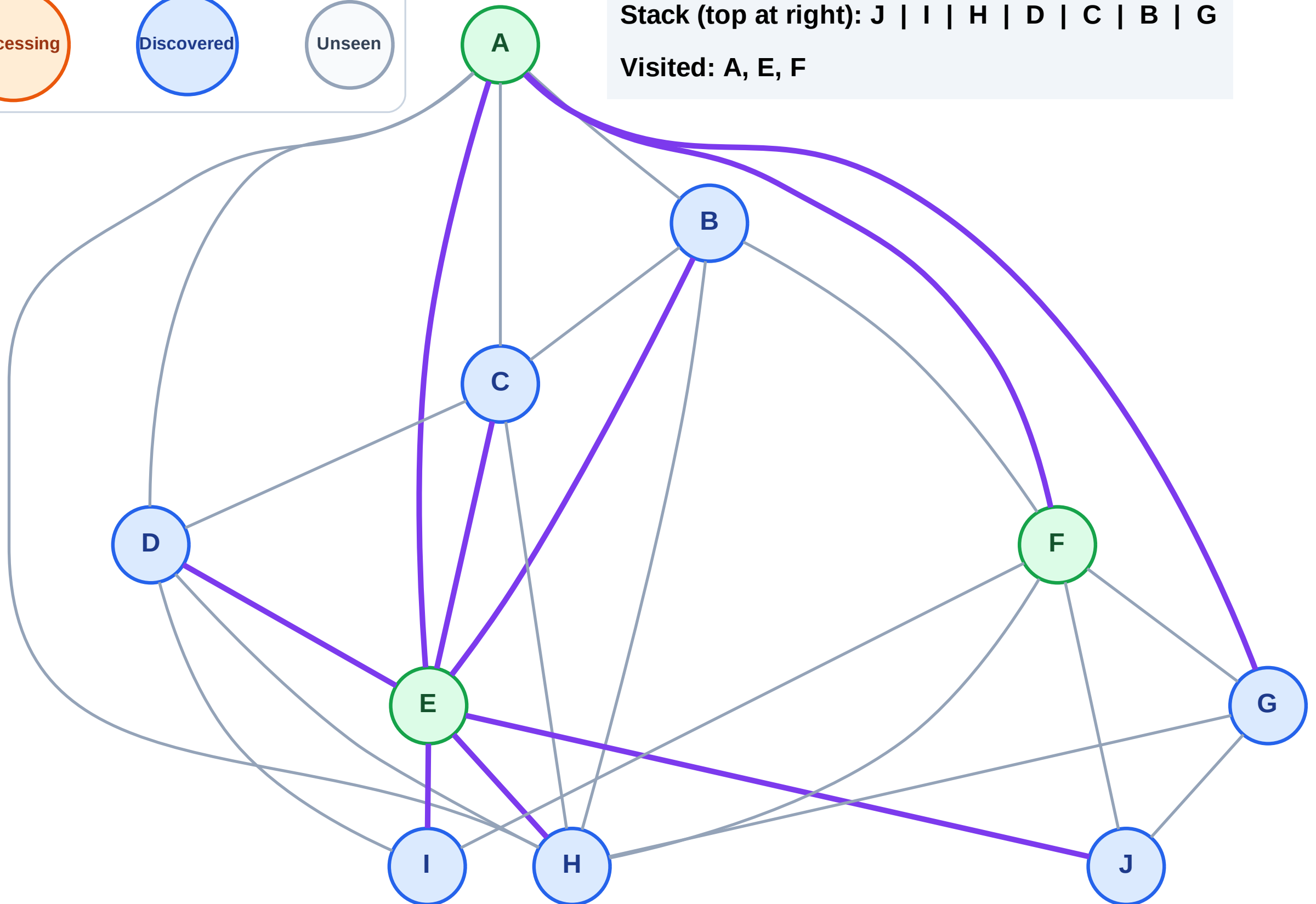
Step 7: No new neighbors from F

Legend



Stack (top at right): J | I | H | D | C | B | G

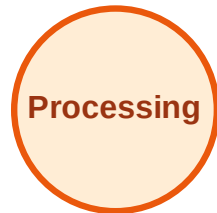
Visited: A, E, F



DFS Traversal

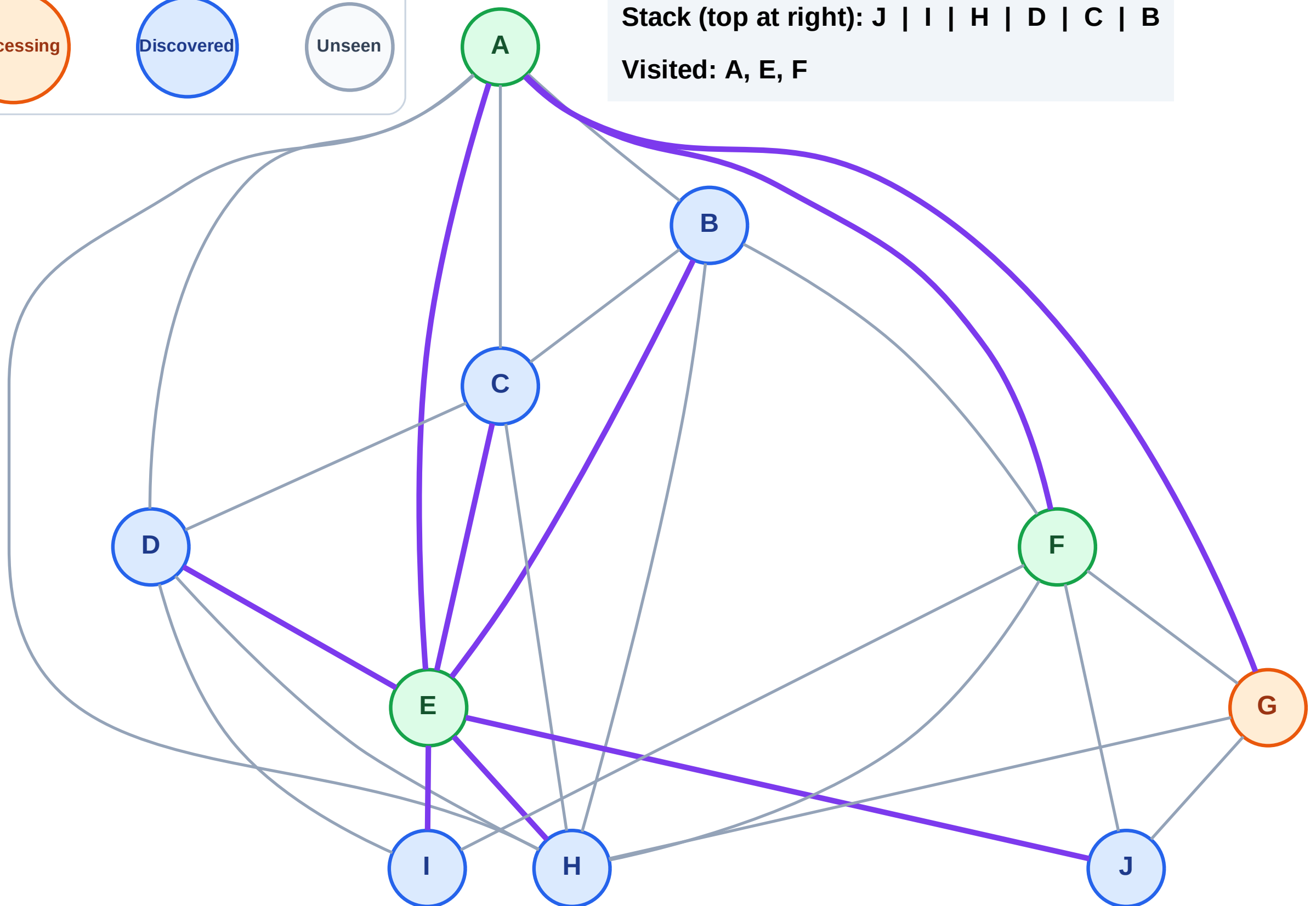
Step 8: Process node G

Legend



Stack (top at right): J | I | H | D | C | B

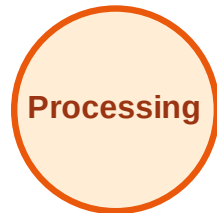
Visited: A, E, F



DFS Traversal

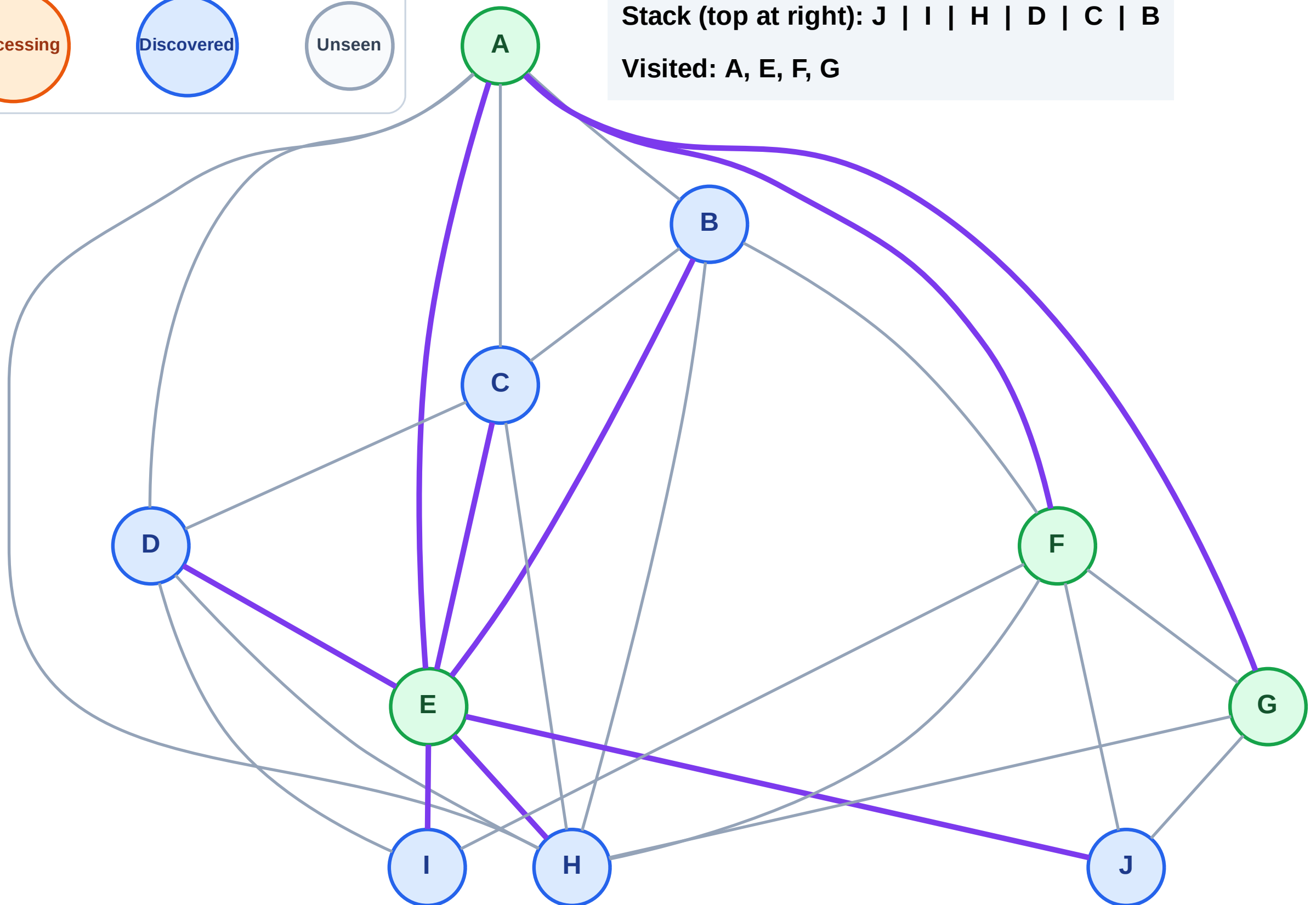
Step 9: No new neighbors from G

Legend



Stack (top at right): J | I | H | D | C | B

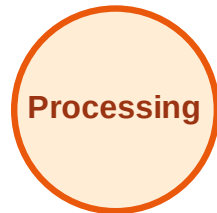
Visited: A, E, F, G



DFS Traversal

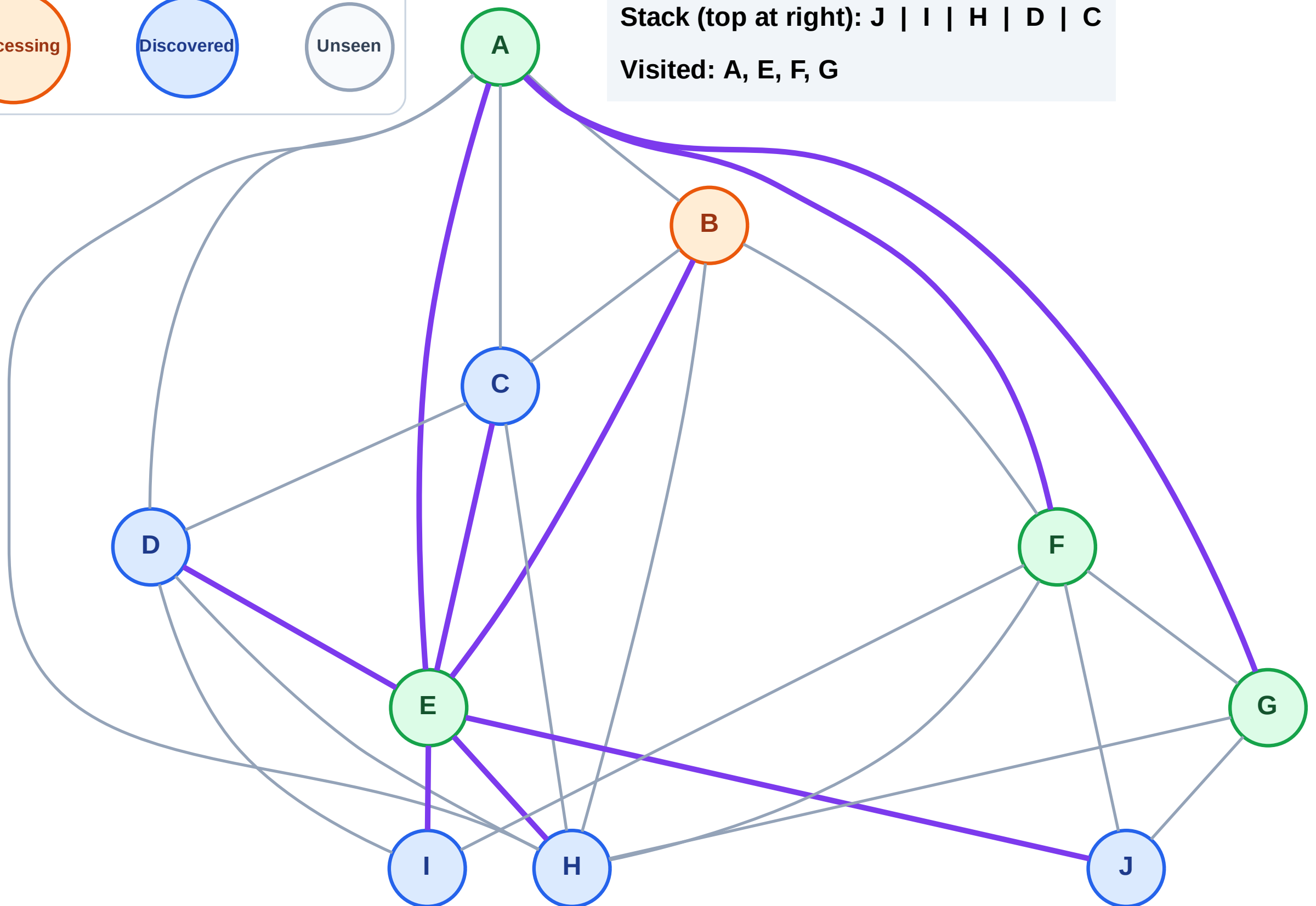
Step 10: Process node B

Legend



Stack (top at right): J | I | H | D | C

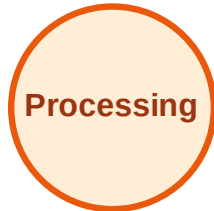
Visited: A, E, F, G



DFS Traversal

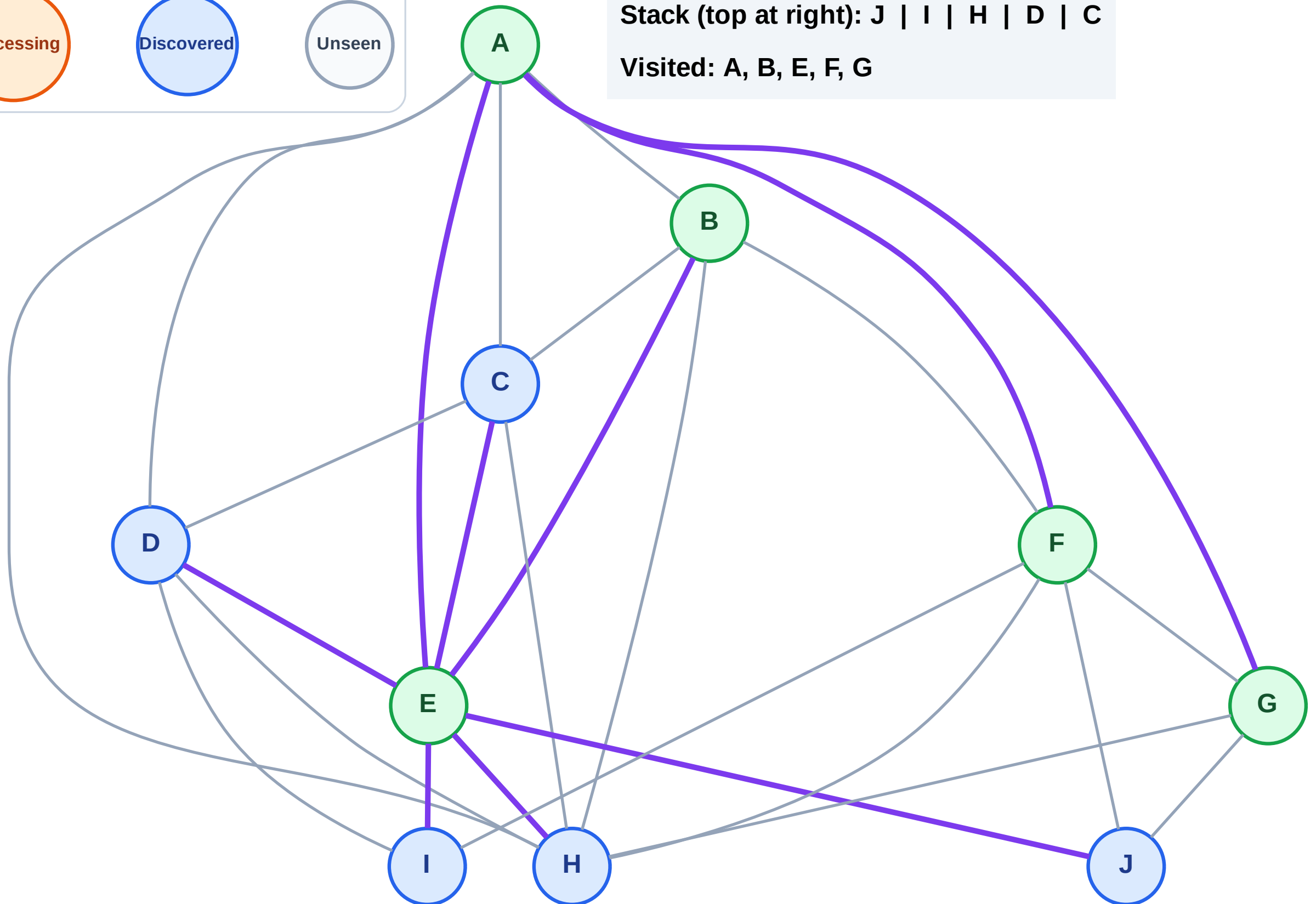
Step 11: No new neighbors from B

Legend



Stack (top at right): J | I | H | D | C

Visited: A, B, E, F, G



DFS Traversal

Step 12: Process node C

Legend

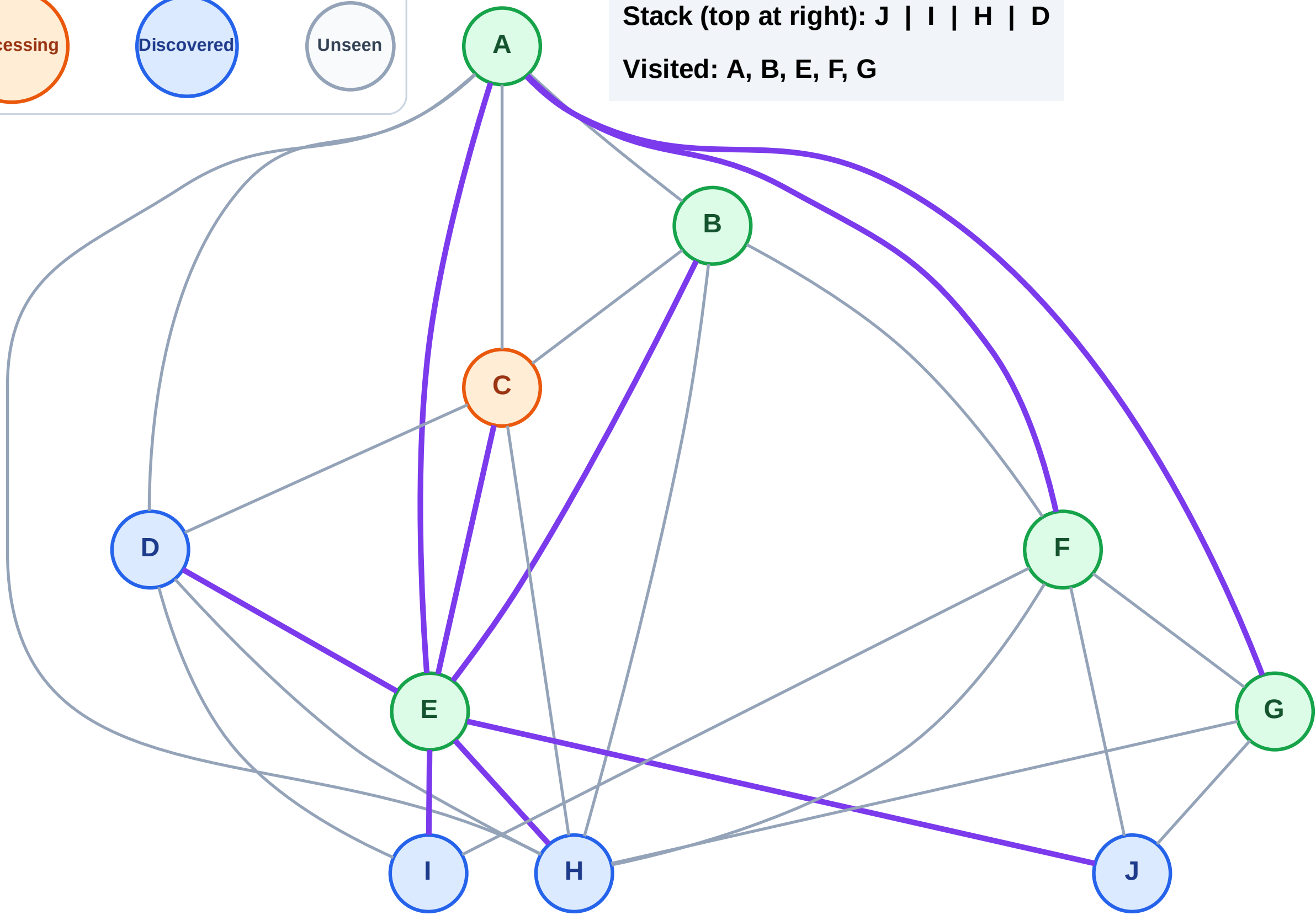
Complete

Processing

Discovered

Unseen

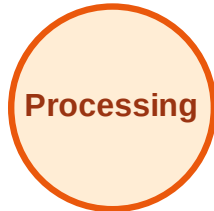
Stack (top at right): J | I | H | D
Visited: A, B, E, F, G



DFS Traversal

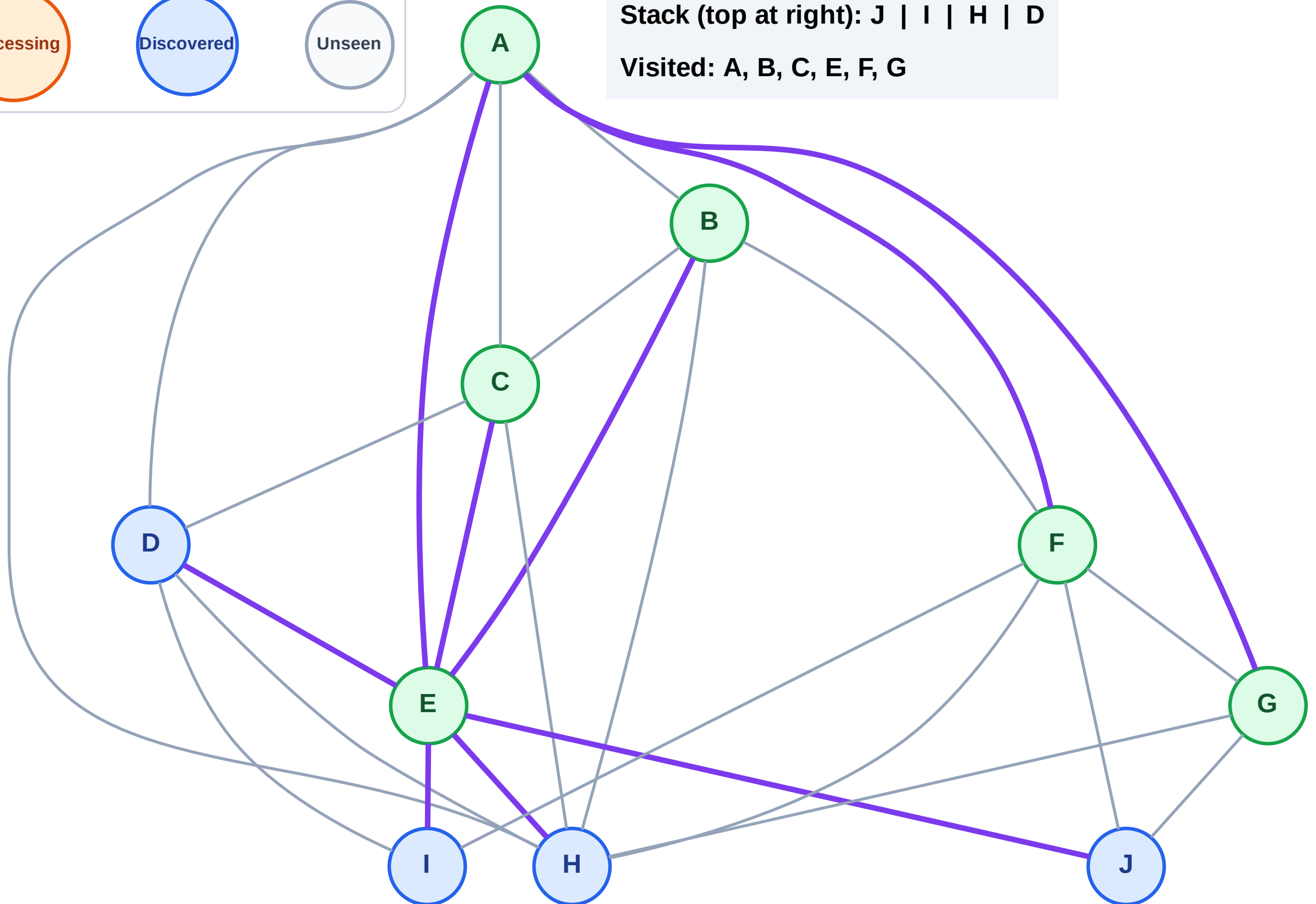
Step 13: No new neighbors from C

Legend



Stack (top at right): J | I | H | D

Visited: A, B, C, E, F, G



DFS Traversal

Step 14: Process node D

Legend

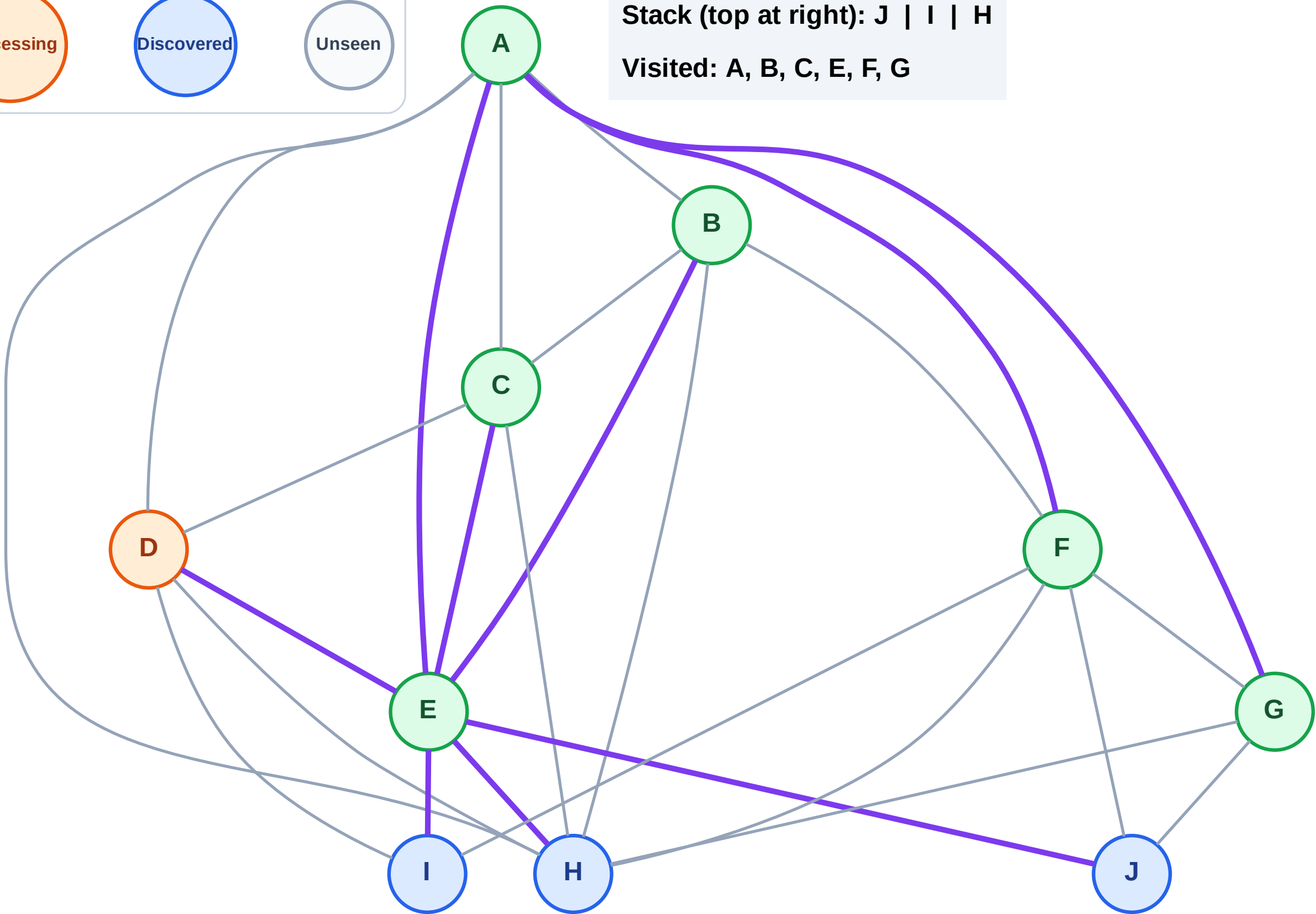
Complete

Processing

Discovered

Unseen

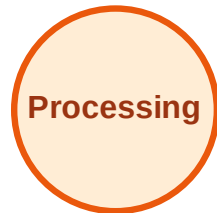
Stack (top at right): J | I | H
Visited: A, B, C, E, F, G



DFS Traversal

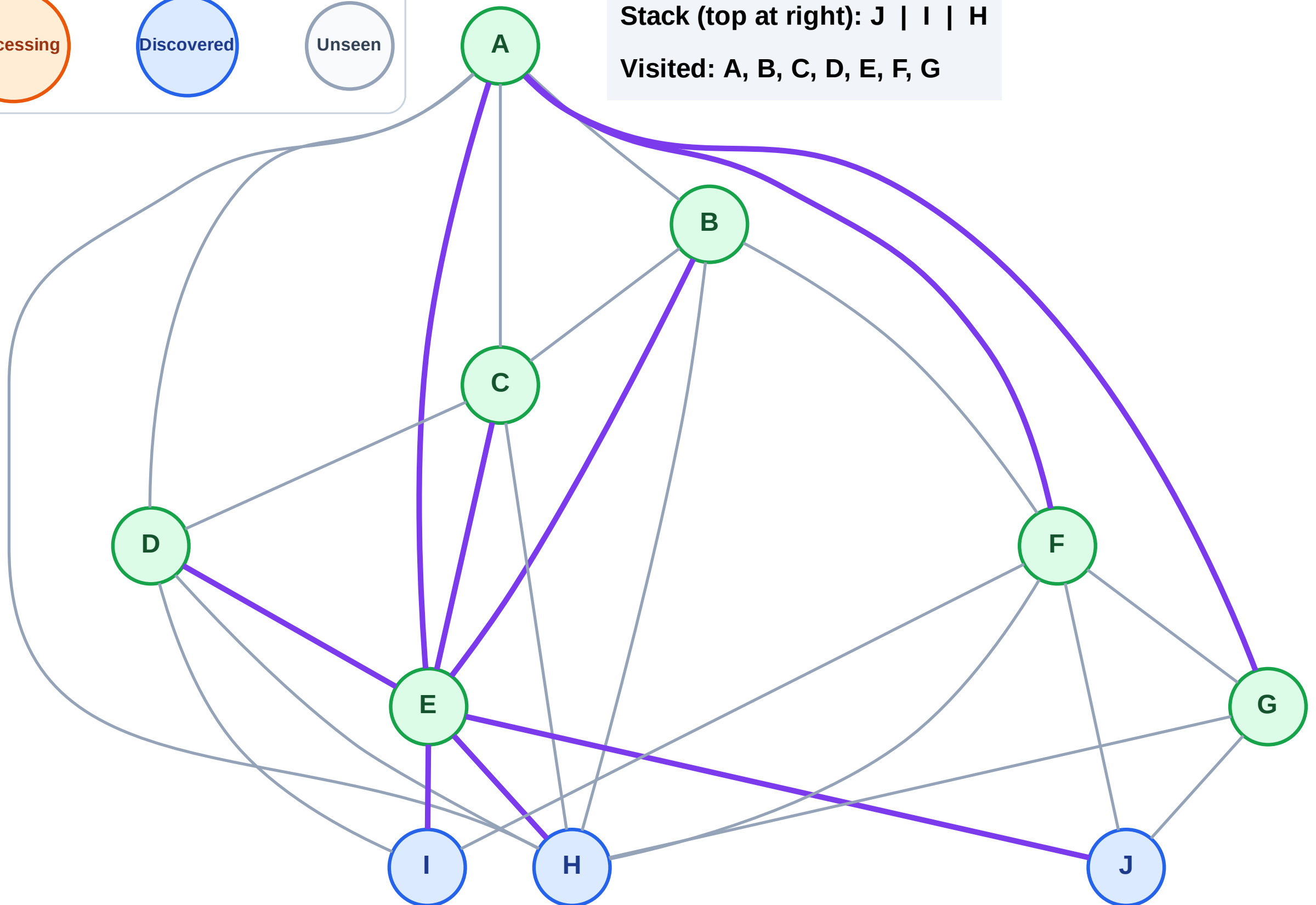
Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | H

Visited: A, B, C, D, E, F, G



DFS Traversal

Step 16: Process node H

Legend

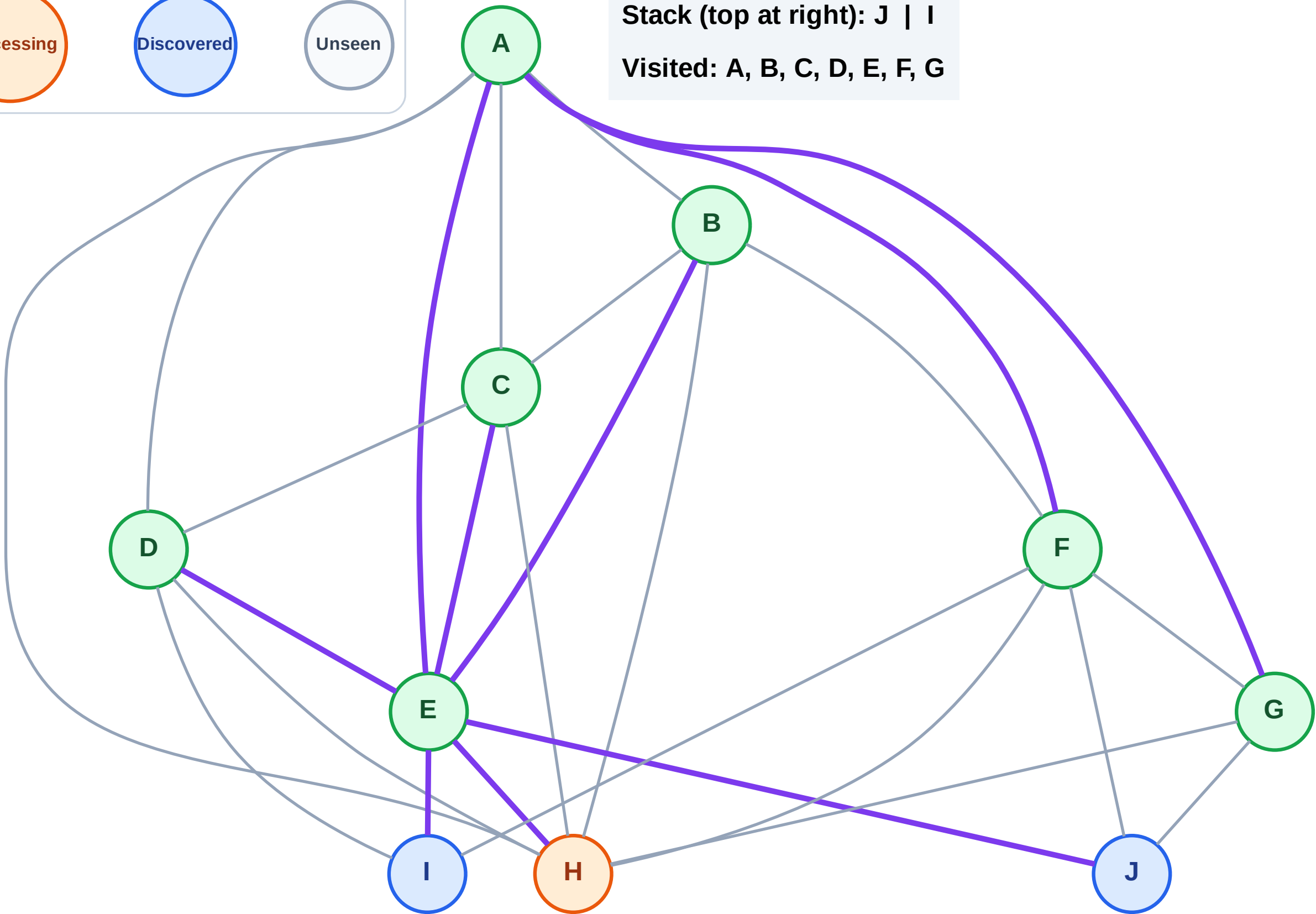
Complete

Processing

Discovered

Unseen

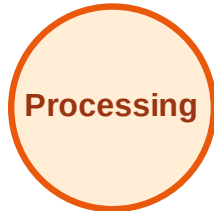
Stack (top at right): J | I
Visited: A, B, C, D, E, F, G



DFS Traversal

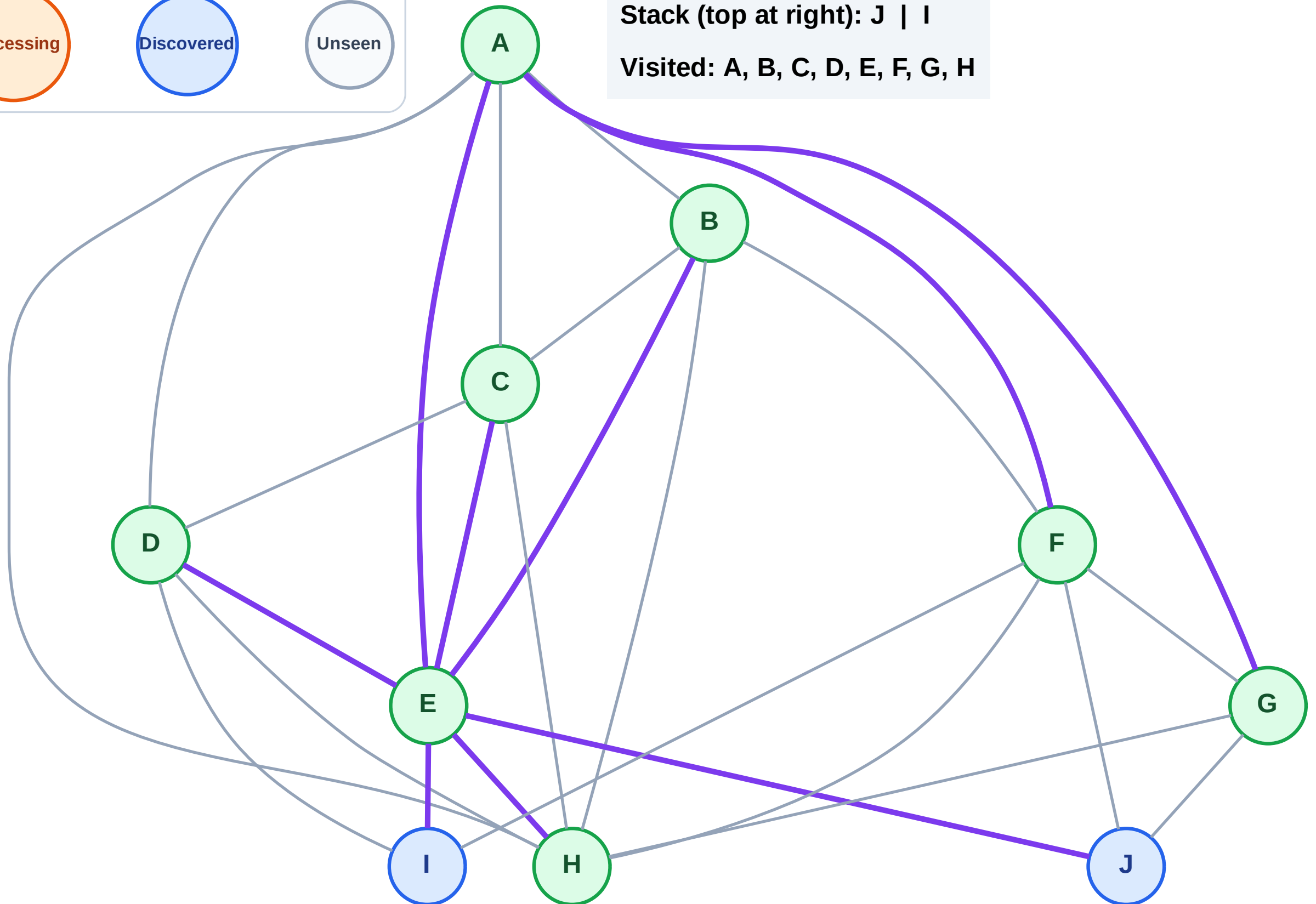
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

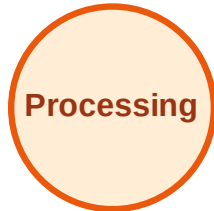
Visited: A, B, C, D, E, F, G, H



DFS Traversal

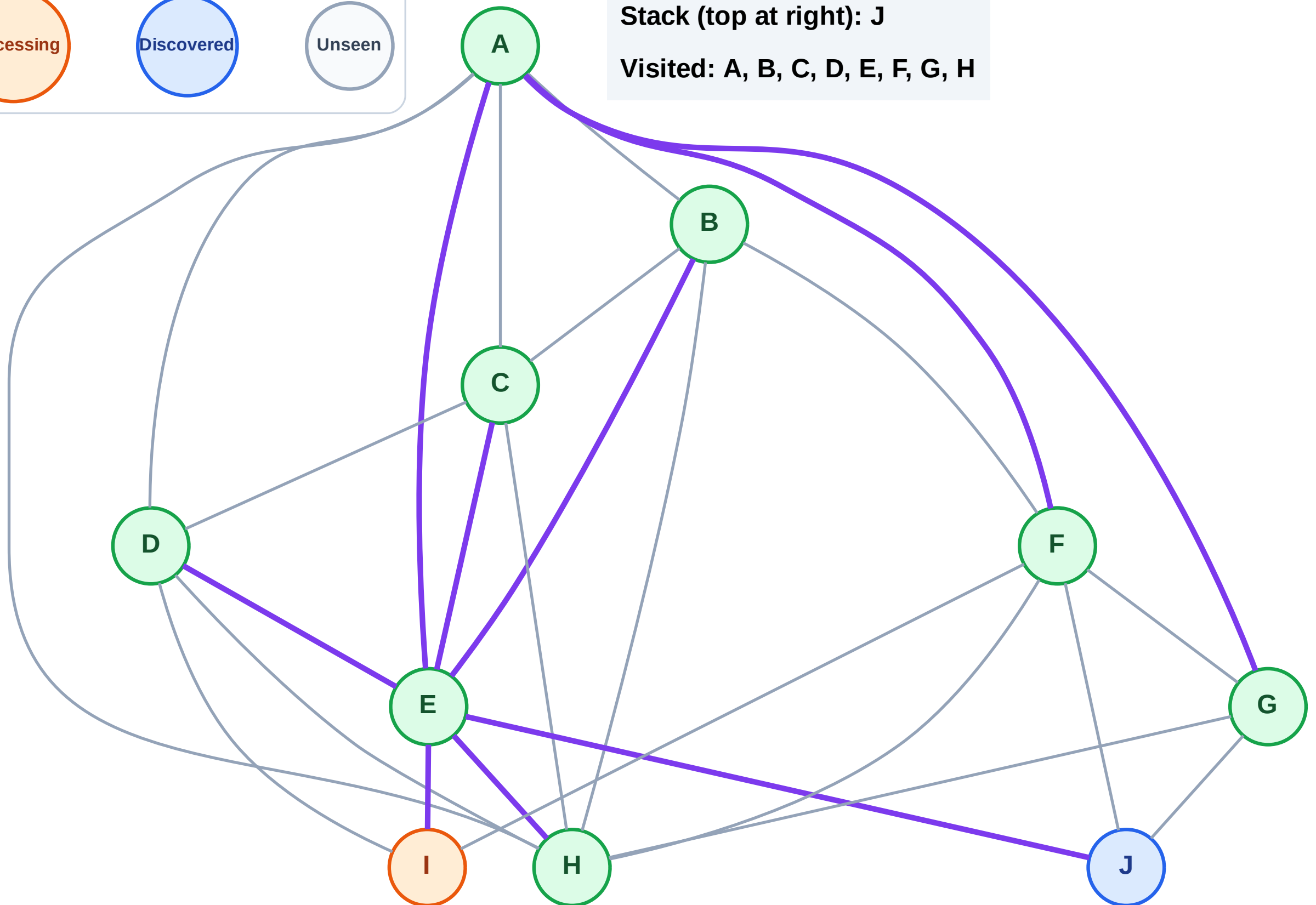
Step 18: Process node I

Legend



Stack (top at right): J

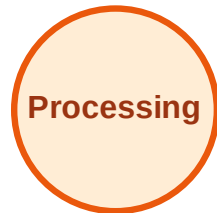
Visited: A, B, C, D, E, F, G, H



DFS Traversal

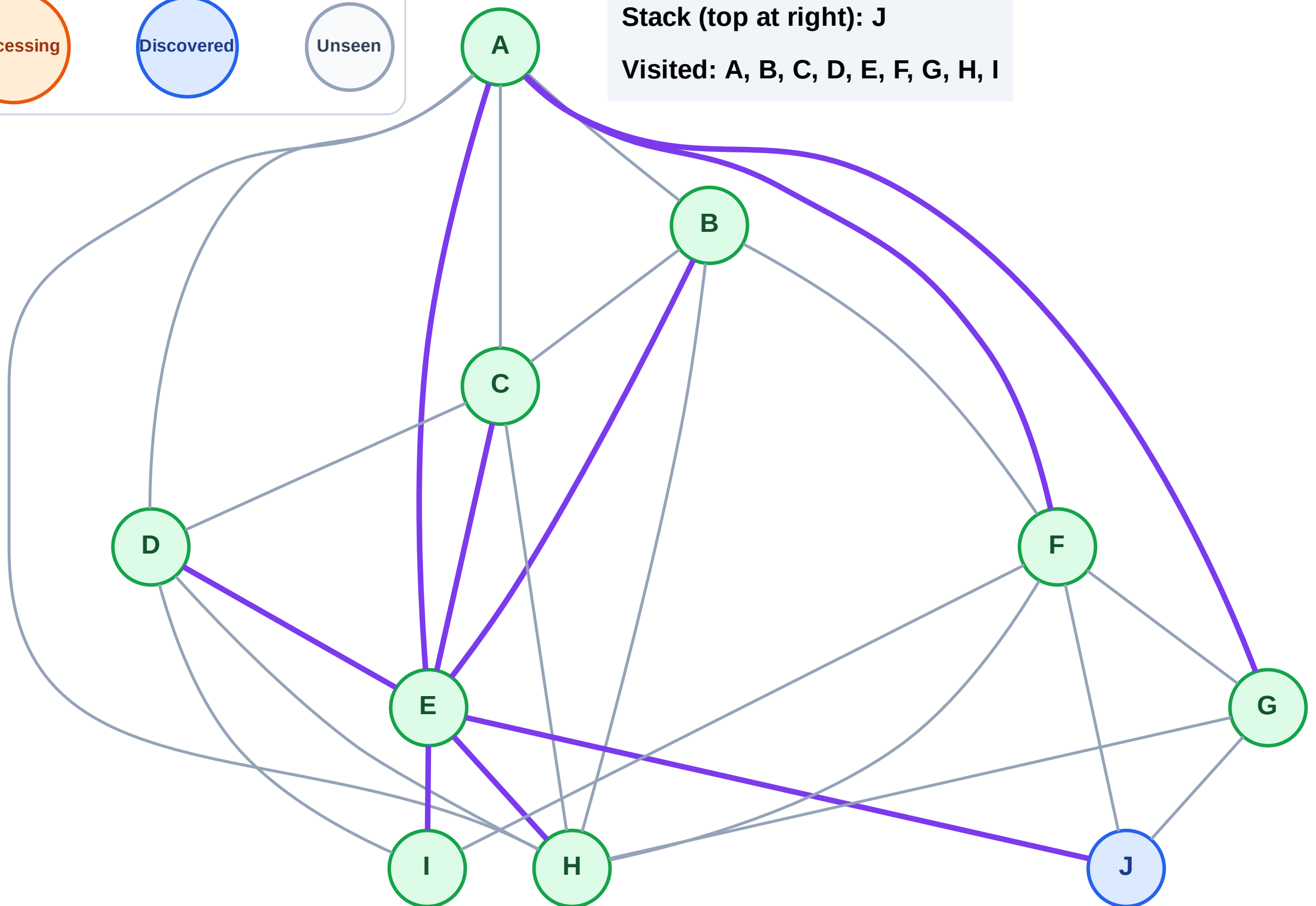
Step 19: No new neighbors from I

Legend



Stack (top at right): J

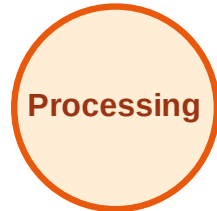
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

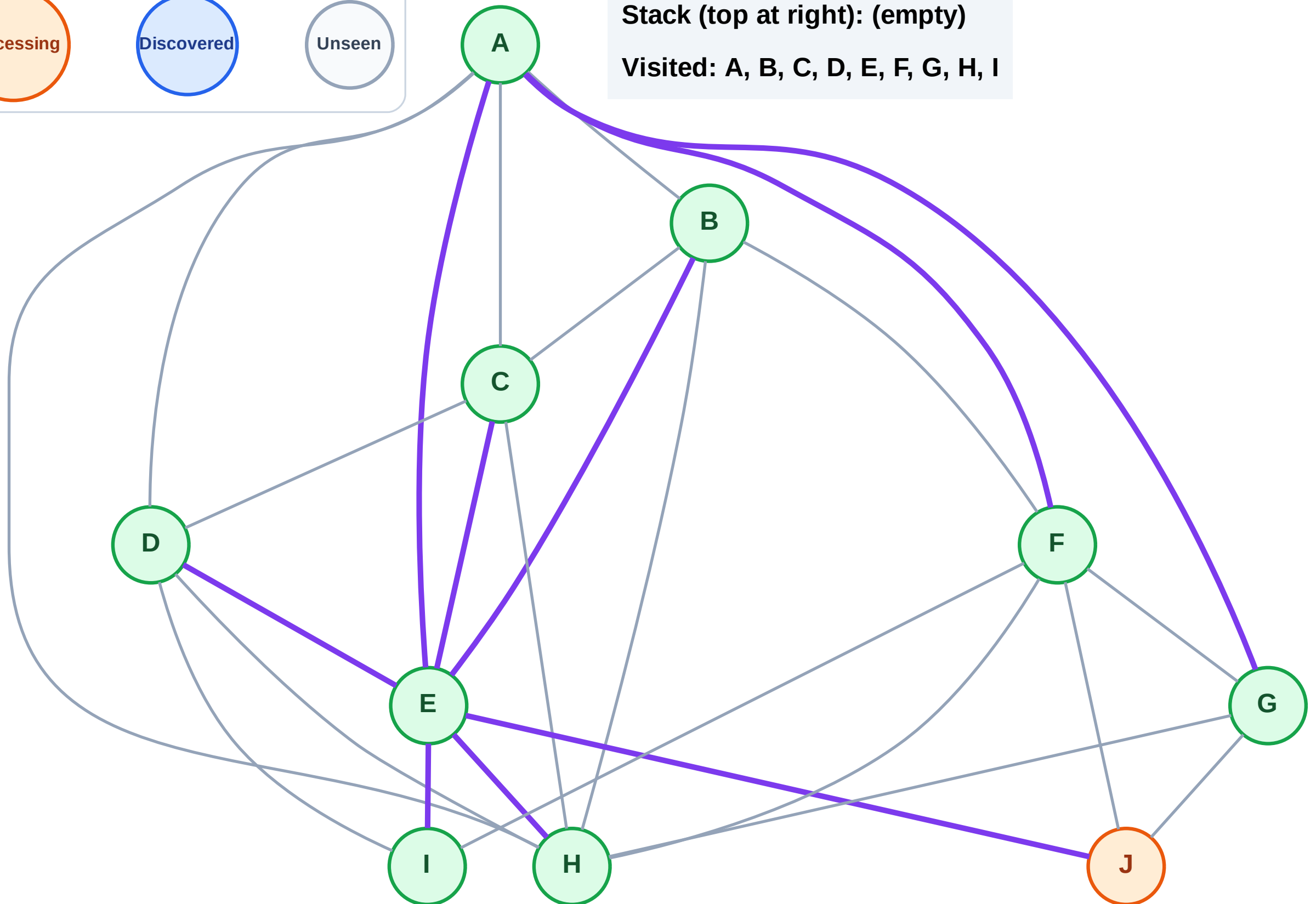
Step 20: Process node J

Legend



Stack (top at right): (empty)

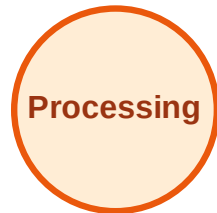
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

